Nest Assured

RFID, ESP-NOW

Google's First Item Tracker.

Never forget your things ever again.

Find out more ⟶

'Google / Google Nest were not engaged in any consultancy or collaborative capacity with this project and the outcome is in no way endorsed by them. Any publicity is limited to personal and academic use'.

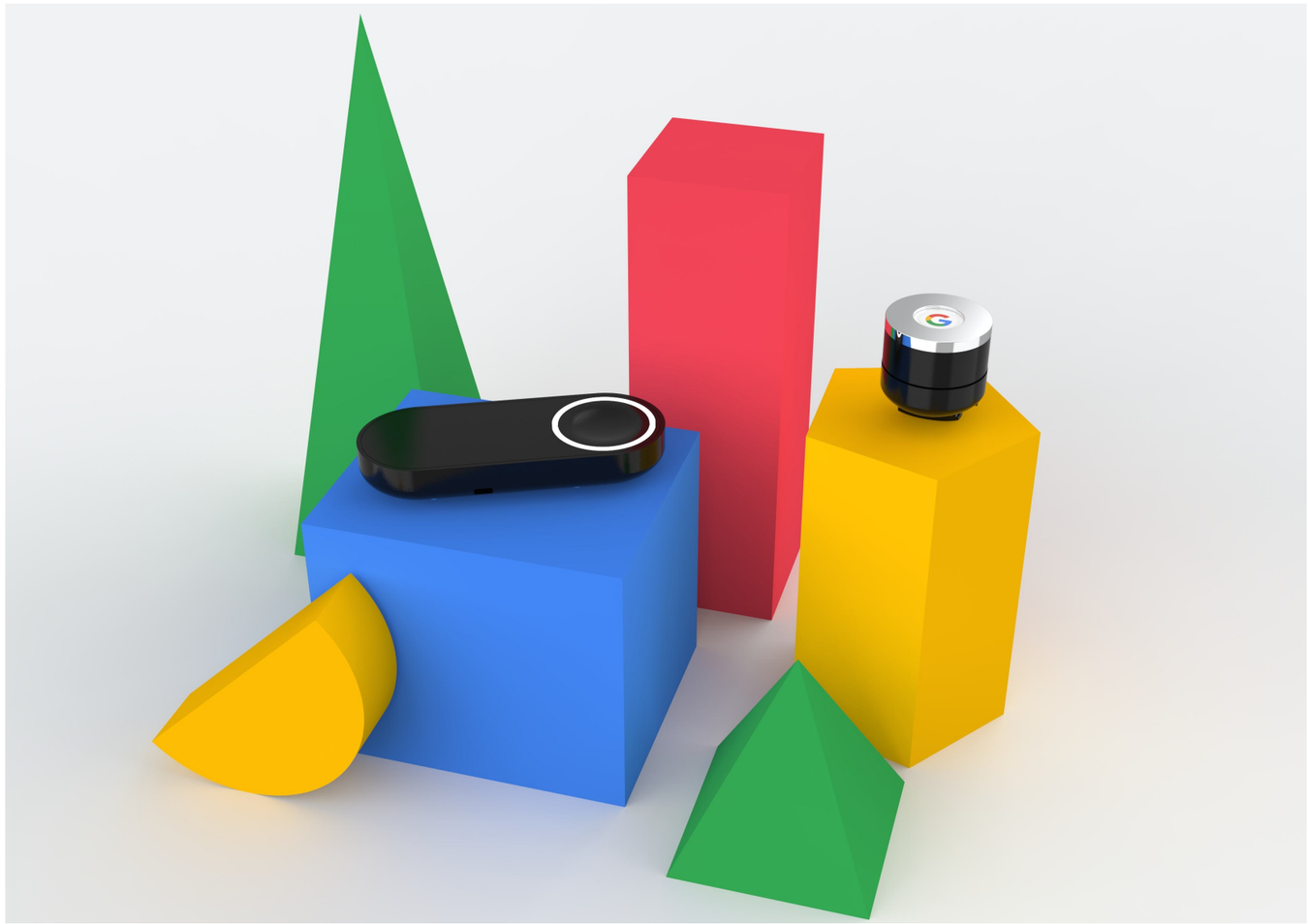
Nest Assured, Rest Assured

- Wireless Communication
- Compact Size
- Easy Integration
- Builds Habits

A plug-and-play tracking solution that integrates seamlessly into dynamic and fast-paced lifestyles. Using RFID and wireless long-range protocol ESPNOW technology, it is compatible with all devices and items.

Designed for young adults with ADHD, but useful to anyone. Nest Assured helps users break free from the endless cycle of over-planning and forgetting.

It is not just another app destined to be lost among others in your phone. It is not another product that demands six other devices from the same brand to function properly.



Secondary Research

ADHD is a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with human functioning or development.

Of adults in the UK have ADHD, equivalent to **1,574,129 people**

It is more prevalent in males than females, with a ratio of 3:1

There are three sub-types of ADHD: Hyperactive-Impusive, Inattentive and Combined. A majority of individuals diagnosed with ADHD fall into the last category.

User Research

ADHD varies between individuals and the symptoms can manifest in a myriad of ways. Interviews allow the individuals to comfortably fully express their thoughts, opinions and feelings. On the same grounds, surveys were not conducted. The reasons for interviews are to

- Allow for variability in responses
- Create a dynamic and personal environment
- Enable ease of discussing personal struggles and experiences
- Allow users to delve into personal experiences comfortably

Selection Rationale

Based on the background research and interviews, a divergent methodology was adopted and 4 initial ideas were ideated.

Pill Dispenser Bottle

Bag Scanner

Smart Bracelet

Task Bracelet

Pill Dispenser Bottle Retro-fitted bottle attachment with compartments for medication storage. Each compartment lights up at the specified time for medication intake.

Bag Scanner Compact clip-on mechanism utilising RFID for item scanning during packing. Provides vibration feedback and progress lights to prevent forgetting essential items.

Smart Bracelet Bracelet monitoring stress levels with a timer for emotional awareness. Accompanied by an app displaying well-being trends over time.

Task Bracelet Community-based bracelet gamifying task completion with friends. Tasks are linked to physical buttons on the bracelet, and completing them collectively earns rewards.

User Feedback & Analysis

The various ideas were analysed against both quantitative and qualitative metrics, and found that Bag Scanner had the greatest development potential. An analysis of market leaders was performed, focusing on the design, functionality, and the gaps. This helped inform the development of the Bag Scanner. The final product will consist of 2 main components.

UHF RFID Scanner

- RFID allows cross-functionality
- Left at bottom of the bag

Controller

- Serves as main user interface
- Controls selection, viewing, scanning

Elegance, durability, and environmental responsibility

Google's home products balance **aesthetics, functionality, and sustainability.**

Using high-quality metals, premium plastics, and sustainably sourced fabrics, the products are **sleek, robust, and long-lasting**.

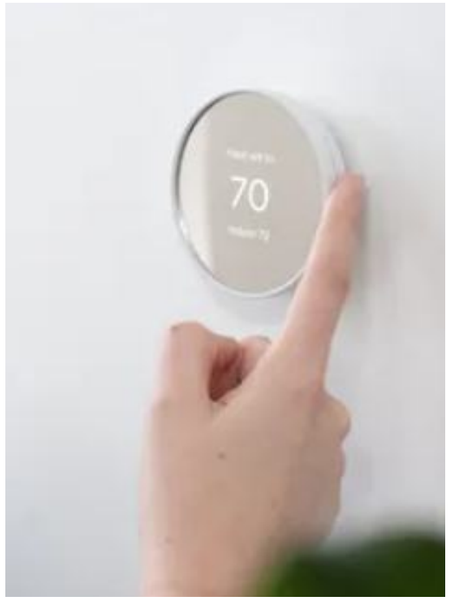
Metals provide a polished, sophisticated look while ensuring sturdiness and longevity. The smooth, cool touch of metal enhances the tactile experience, adding to the premium feel.

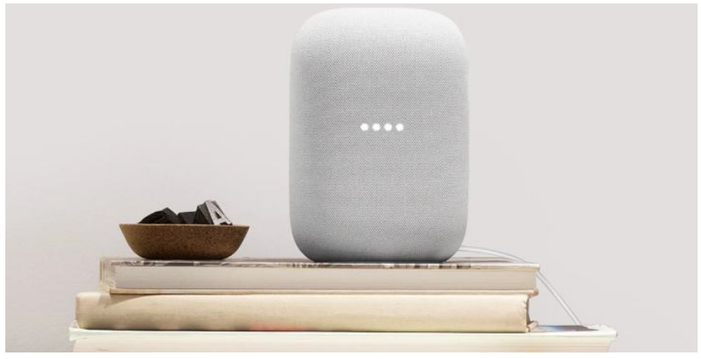
Premium plastics are chosen for their versatility and ergonomic design potential, often treated for a soft-touch feel to enhance comfort and grip.

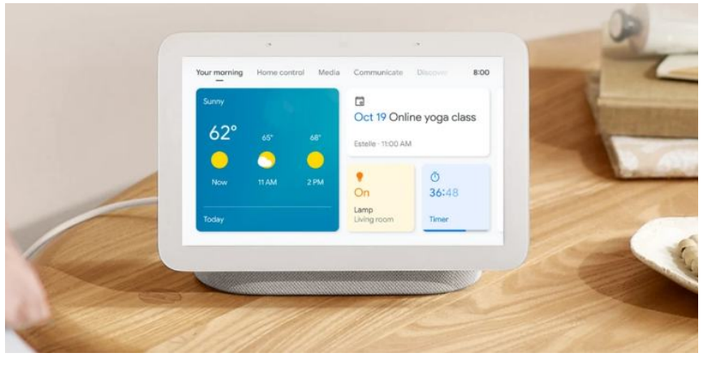
Sustainability is a core consideration, with Google incorporating recycled materials like plastics and metals, as well as sustainably sourced fabrics. This commitment ensures materials are safe, environmentally friendly, and contribute to a circular economy.

Comparisons to Google's Smart Home products gave insights on their design, features and general aesthetics.

Experiencing their designs firsthand gave us a clear understanding of the finishes we needed to achieve.







Concept Development

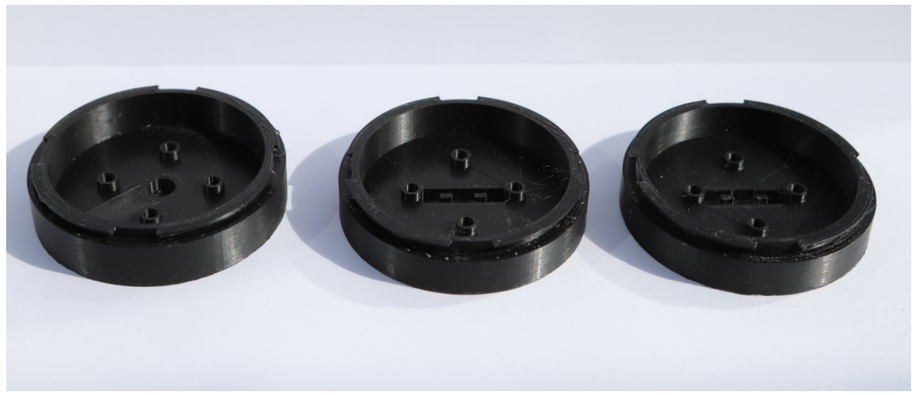
Nest Assured's design is inspired by and rooted in Google's strive create a home that looks after the people inside it.

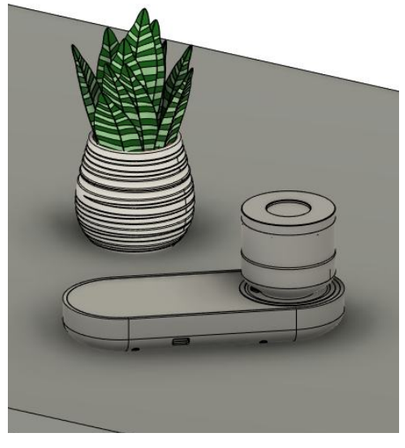
Compact size
Keeping the size of the device minimal was a key focus. Reducing any wasted space while achieving functional integration between components required careful planning.

Tactile and comfortable
It is crucial for the dial to have unmatched ergonomics and interactions. Initial prototypes made from cardboard and tape developed into 3D printed samples to test size and shape

Aesthetic integration
Although the two parts will be placed in different parts of the bag, keeping the lines, shapes, colors and textures the same really makes them stand out as integrated devices.

User testing
By interviewing the 6 users contacted in the previous user phase and conducting collaboration sessions, Nest Assured physical and digital design was validated.





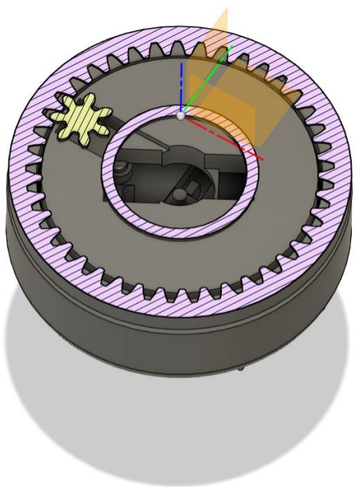
Physical Hurdles

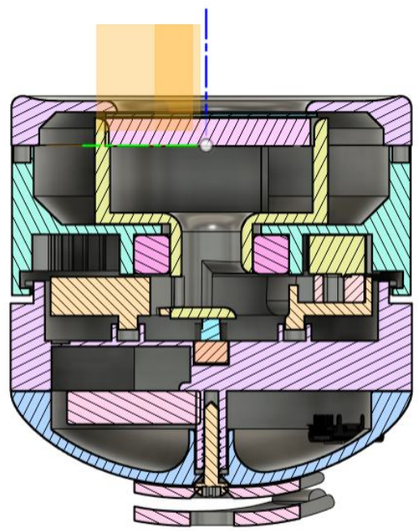
Google's Nest products are designed to be portable and unobstrusive, ensuring they fit seamlessly into various living environments. Multiple CAD iterations were developed to compactly integrate the internal components into an even smaller form.

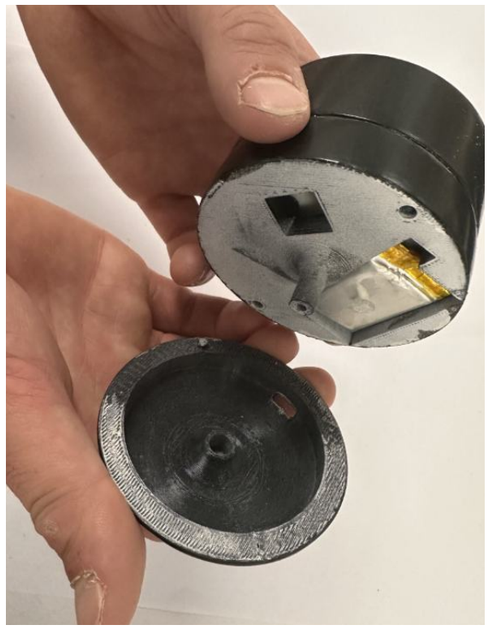
One of the key issues we faced with designing the device was maintaining the bottom and screen stationary, whilst allowing the top dial to rotate. An internal, hollow, stationary shaft was designed to connect the bottom casing and the screen,

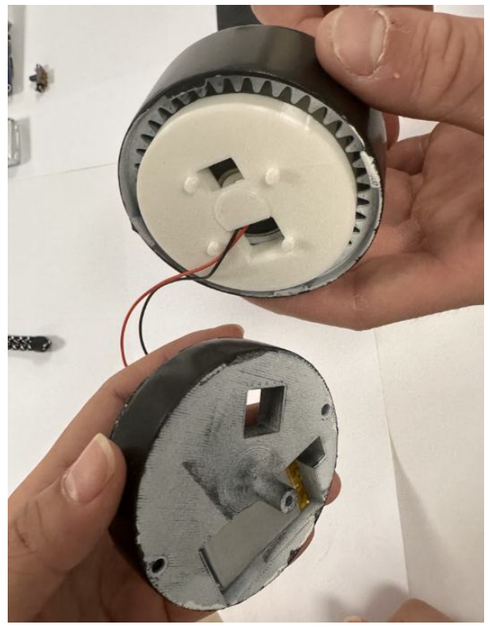
Another critical challenge was allowing space for all of the electrical connections. The most charging wire in particular which runs from the screen to the bottom of the dial was the most intrusive as it uses a much thicker cable than the other electronics.

The section analysis tool was useful to see the inner workings.







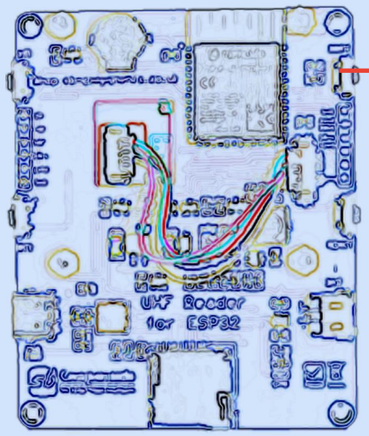


Nest

Power, Component and Load Specifications

Standardised Components:

System Control:



UHF RFID ESP32 Module

Up to 1.5m range

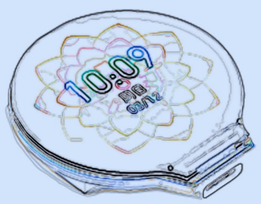
Multiple read - up to 50 tags per second

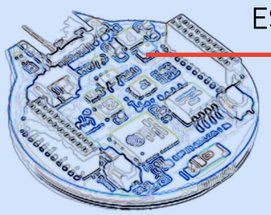
Operating pin voltage 3.3V

Board supply 5V

USBC and Battery plugs

Communication with ESP NOW



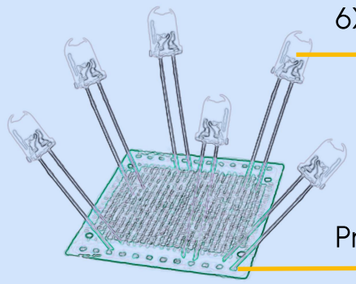


ESP32 LCD Screen 1.28″

3.5V battery supply required

Built in Accelerometer and Gyroscope

Base Components:



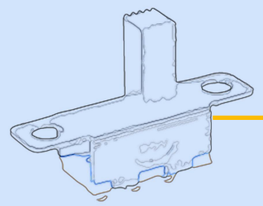
6X White LED

3.3V, 0.02A

Connected to individual GPIO pins

Prototype Board

Would be a custom PCB in a production run

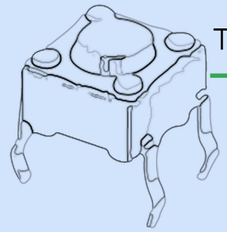


Slide Switch

<0.01A, 3.3V

Used to turn the baseplate on and off

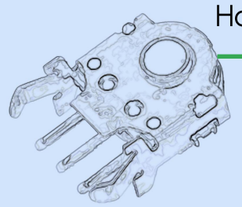
Dial Components:



Tactile switch 6mm

<0.01A, 3.3V

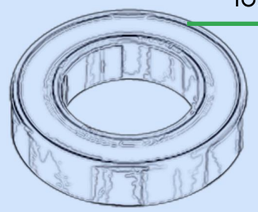
7.5KOhm pullup resistors



Hollow Shaft Rotary Encoder

<0.01A, 3.3V

A pin = direction 1
B pin = direction 2
C pin = Ground



15X25X7mm Bearing

Ensures consistent smooth turning of the dial.

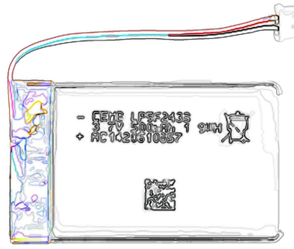
Power Management:

Base Battery

3.7V 2000mAh Lipo attery

Built in BMS

Higher storage capacity than the screen as there are more intensive components

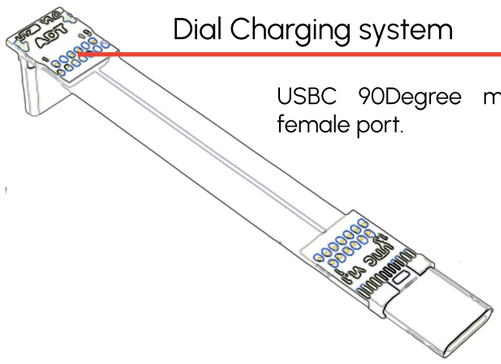


Dial Battery

3.7V 500mAh Lipo battery

Built in BMS

Recommended capacity from screen manufacturer.

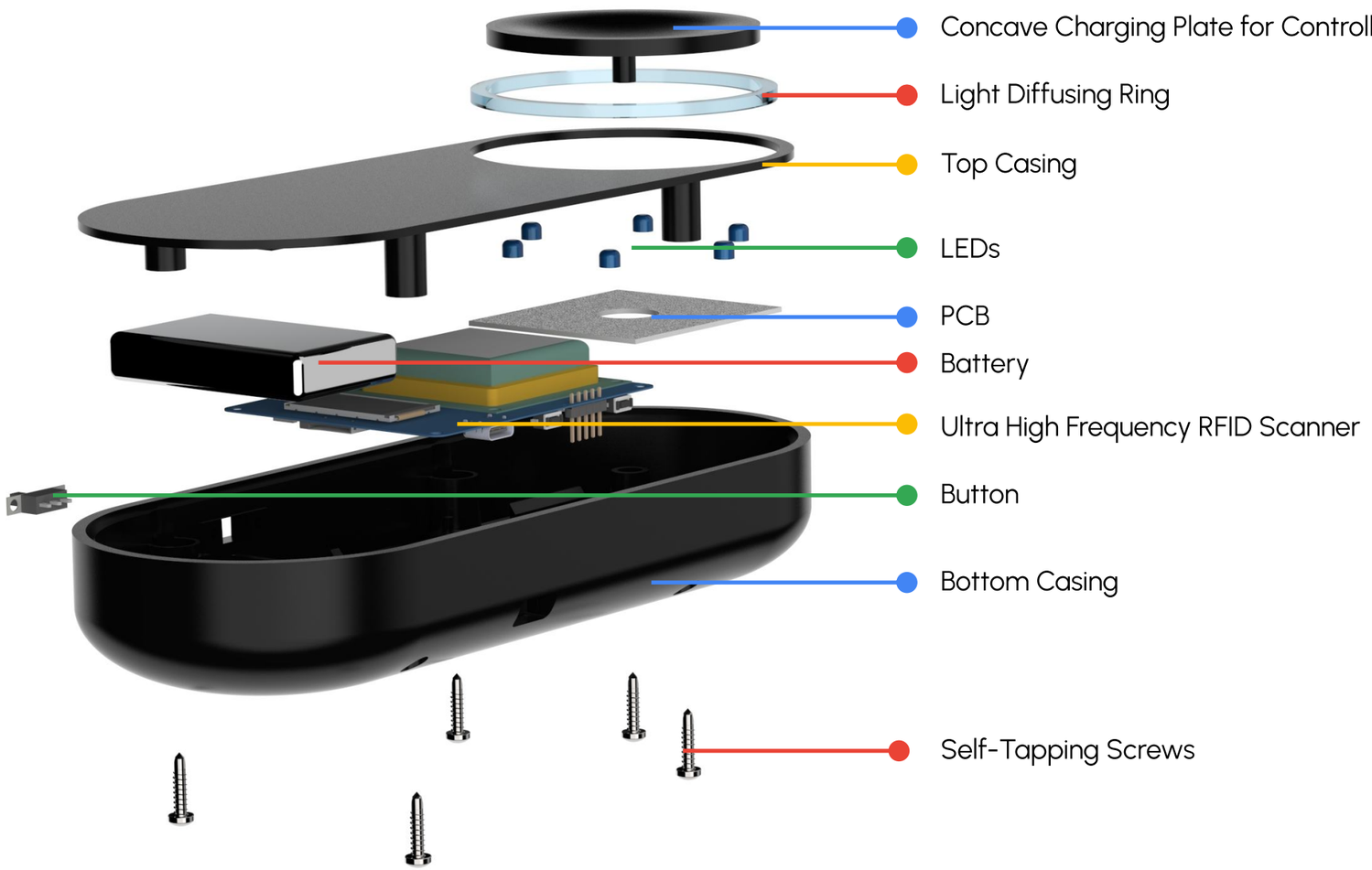


Dial Charging system

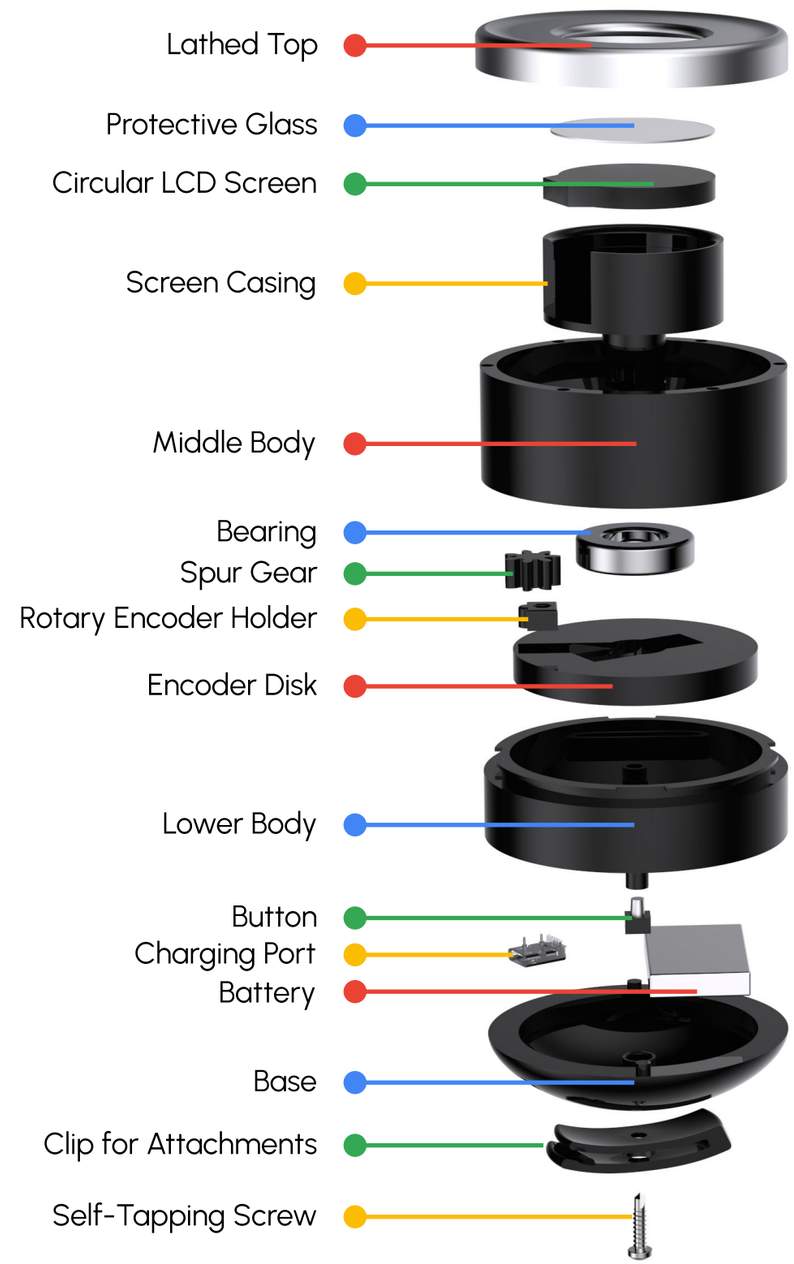
USBC 90Degree male, to female port.

Estimated recharge interval: Once a week

Scanner



Controller



Ergonomic Concave Dial

Lathed with aluminium, the smooth round dial provides tactile comfort, functional precision and premium feel, instigating further interaction with the device.

Seamless

By incorporating ribs, grooves, and raised edges within the product, the material remains hidden from sight. This seamless design merges the components into a single, unified product.

Tactile feedback

The effortless adjustment mechanism offers immediate feedback, creating a harmonious and streamlined interaction.

Ergonomic Comfort

The smooth, curved surface ensures a comfortable hold, offering a natural grip and reducing pressure points for the user.

Multi-functional attachment

The interchangeable, customizable attachment enhances the design, offering a secure and visible method for transporting the product while on the move.

Clear Glass

The luminescent ring serves as a clear visual indicator of where to position the product, providing an sophisticated look and an aesthetic sensory integration.

Polished Aluminium

The metal bezel elegantly frames the screen, enhancing the picture with a seamless extension. Its cool-to-touch surface and sleek, contemporary appearance add sophistication and convey a premium feel. This aligns perfectly with Google's commitment to quality and durability.

Premium Plastic

The premium material ensures a durable, impact resistant, consistent look across the product, providing a lightweight and comfortable experience

Glossy Coating

The black coating provides an extra layer of protection and a smooth, shiny finish which enhances the quality of the product and grounds the product in Google's Smart Home line-up.

Enhancing everyday life

'At Google Nest, our mission is to create a home that takes care of the people inside it and the world around it. For us, the home of the future is not a smart home, it's a helpful one.'

Nest Assured

£99.99

6W x 12H x 7.2D cm

"Be prepared. Stress less"

Compact	Timeless	Resistant
Aluminium	Black matte	Rubber coating

High end products are lightweight yet sturdy, ensuring it can withstand regular use and maintain its aesthetic appeal over time.

Nest Cam

£89.99

7.28W x 11.45H x 6.92D cm

"Know more. Worry less"

Protective	Sleek	Reliable
Rubber coating	Black matte	Technology

"Set yourself apart from competitors"

Nest Doorbell

£179.99

4.6W x 16H x 2.4D cm

"Know more. No wires required"

Glowing	Smooth	Modern
Light ring	High-grade plastic	Speaker

"Your helpful home starts here"

Nest Thermostat

£219

4.2W x 4.2H x 3.08D cm

"It's easy to save energy"

Metallic	Crisp	Sturdy
Bezel	Glass	High-grade plastic

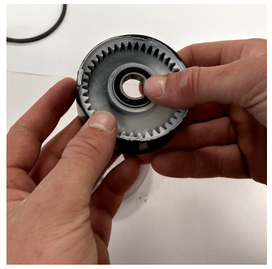
Home products are generally small, portable and compact

Google Environment

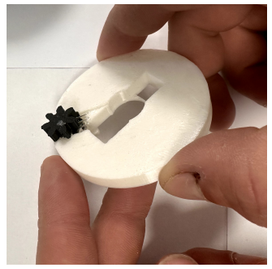
After rendering our final product we integrated it into the environment alongside other Google product ranges.

- Nest Mini
- Nest Audio
- Nest Thermostat
- **Nest Assured**
- Nest Doorbell
- Pixel Tablet
- Nest Cam

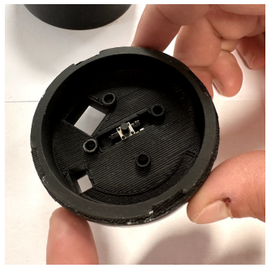
Controller



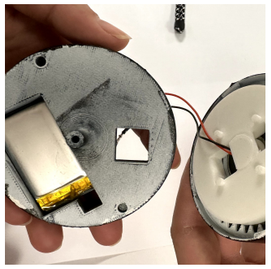
Press bearing into middle body



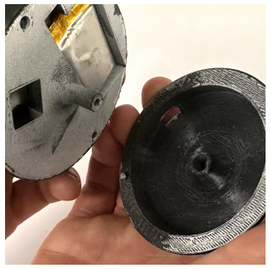
Place rotary encoder onto holder



Attach button to underside of lower body



Attach battery and feed cable through



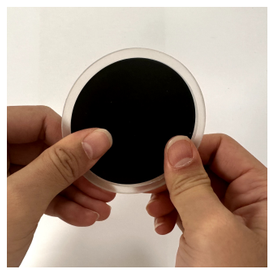
Attach base

Screw clip-on onto base

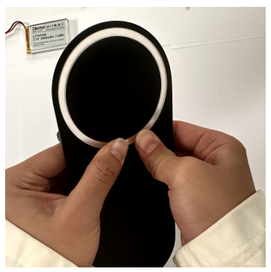
Attach screen to the holder

Attach lathed top

Scanner



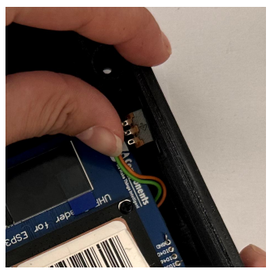
Push charging plate through ring



Place both parts through the top cover

Lay UHF RFID Reader and push PCB into snap fits

Attach battery and connect



Feed power button through cut-out slot

Add cotton gauze for light diffusion

Place lid

Screw close the case

Total assembly time:

14 mins 29 seconds

Total assembly cost:

$1.70 USD

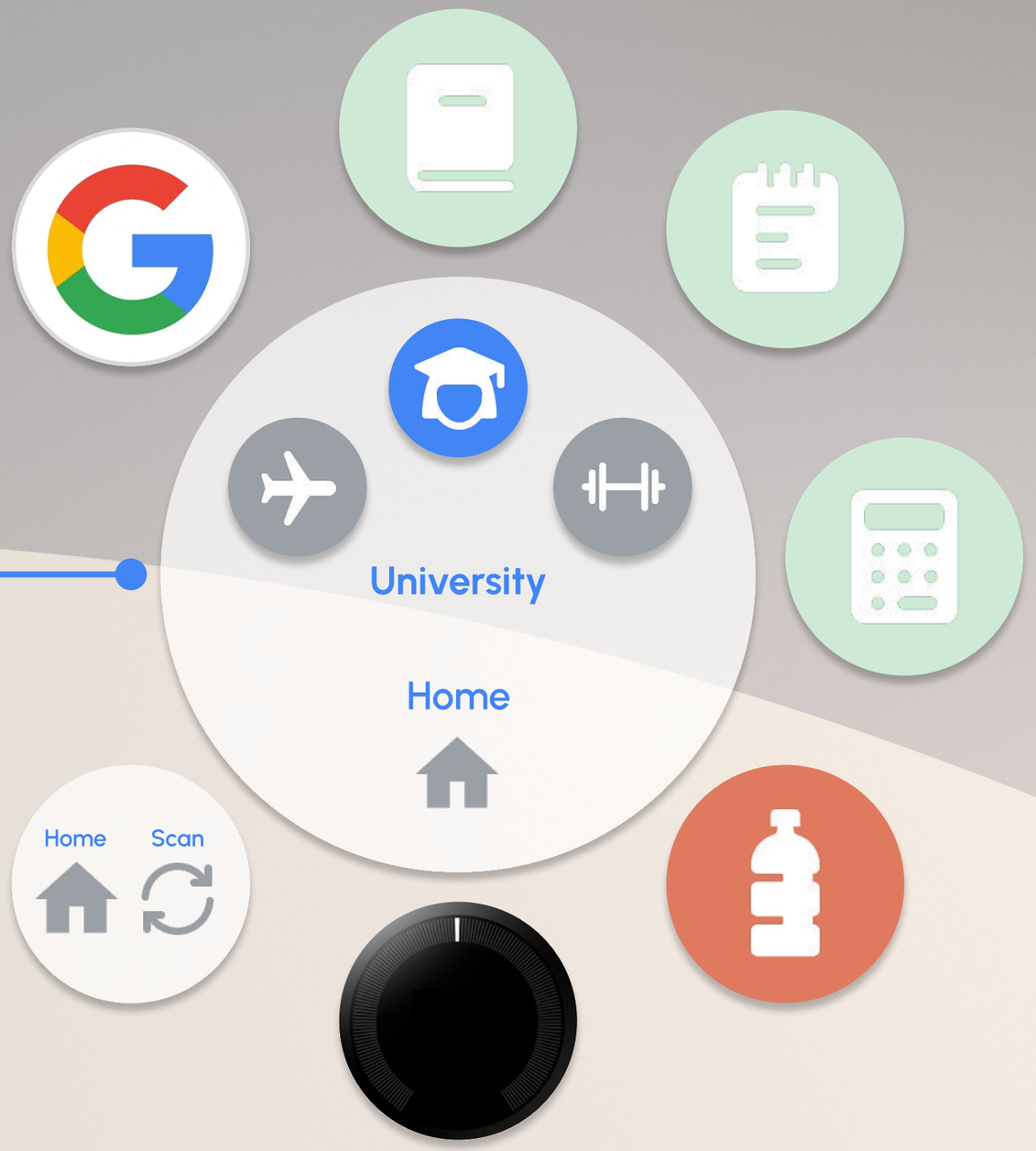
Based on 7 USD per hour rate

Intuitive

The UI is designed to be simple and clean, minimizing distractions for users. Icons are used instead of text, making the interface clearer, especially for young people with ADHD who can be easily overwhelmed.

Effortless

The key design feature of the UI and dial is its one-hand operability, allowing navigation through just two interactions: rotating and pressing the dial.



Packaging Compliance

Relevant standards added to the packing

Actual product color may vary from the image

Imported into the EU by
Google Commerce Limited
70 Sir John Rogerson Quay, Dublin 2, Ireland

Designed by Google. Made in Thailand.

Safety Guide

CAUTION!

Do not expose Nest Assured to liquids.

It can cause damage to the electronic components

HEALTH AND SAFETY

- Read these instructions
- Keep these instructions
- Heed all warnings
- Follow all instructions
- Do not expose device to liquids
- Do not disassemble, crop, band, burn, crush or puncture device
- Do not use device if screen is cracked or broken
- If it gets wet, do not attempt to dry it using an external heat source
- Do not use device in the bathroom

- Only use attachments/accessories specified by the manufacturer
- Refer all servicing to qualified personnel. Servicing is required when the device has been damaged in any way, such as power-supply cor or screen is damaged, liquid has been spilled or objects have fallen into the apparatus, the apparatus has been exposed to the rain or moisture, does not operate normally or has been dropped.
- WARNING: To reduce risk of fire or electric shock, do not expose device to rain or moisture. Do not expose this equipment to dripping or splashing and ensure objects filled with liquids, such as water bottles, are correctly sealed.

Electrical Shock Hazard Symbol warns users of the risk of electric shock if the product is mishandled or if protective measures are not followed.

Exclamation point within an equilateral triangle alerts users to the presence of important operating maintenance (servicing) instructions in the manual accompanying the product.

The product can be turned off completely by switching button off. Mains socket must always be accessible and unobstructed to charge

Electrical and electronic equipment, ports and batteries marked with the WEEE symbol must be collected and disposed of separately to protect the environment.
Google retailer will advise you of the correct disposal method in your country.

IMPORTANT!

- Do not expose device to direct sunlight.
- Make sure device is set up, placed and connected in accordance with the instructions.
- Do not place heavy items on top of your product.
- Do not subject the product to high humidity, rain or sources of heat.
- The product is designed for indoor use in dry environments only. Use within a temperature range of 0-25°C

Additional Support

Google

Nest Assured

Additional information at:
g.co/nestassured/support

Care And Cleaning

- Unplug device and adaptor before cleaning
- Unplug device during lightning storms
- Unplug device when unused for long periods of time
- Use clean soft, dry cloth to clean device
- Don’t use any chemical detergent, powder or other chemical agents (such as alcohol or benzene) to clean device

Product Labelling

MODEL N0776ST. ASSEMBLED IN THAILAND.

Google 70 Sir John Rogerson Quay, Dublin 2, Ireland

Detailed User Guide

Online user guide offers detailed troubleshooting information and can be accessed using the URL code provided on the packaging

Eco-friendly

While Google's standard packaging reflects the company's broader commitment to environmental responsibility, **challenges remain in achieving full sustainability due to supply chain complexities, the use of mixed materials, and the need for better recycling information.**

Standard Packaging

Google typically features a clean, minimalist design with a focus on simplicity and elegance

Color Scheme
White or light background with Google's signature colours

Material
Boxes made from high-quality, sturdy cardboard.

Branding
Google logo and product name is displayed on the front.

Image
High-quality image of the product on the front of the box.

Text
Minimal, clean text, focusing on product name and key features.

Organisation
Well-organised, with compartments and protective inserts.

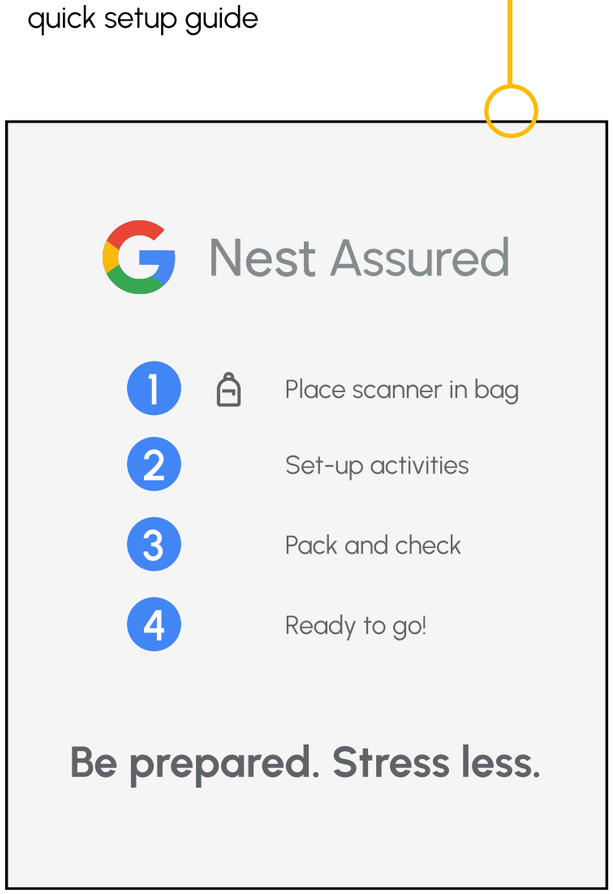
Included Manual

Manual provides quick start guide and safety guidelines

Sustainable packaging

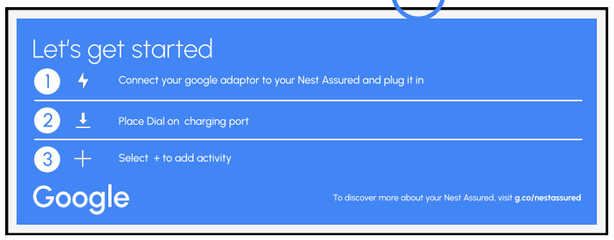
Integrated User Guide

Internals of packaging display the quick setup guide



Printed Manual

Start guide is printed on the side for clear, visual guidance



Mechanical Sealing

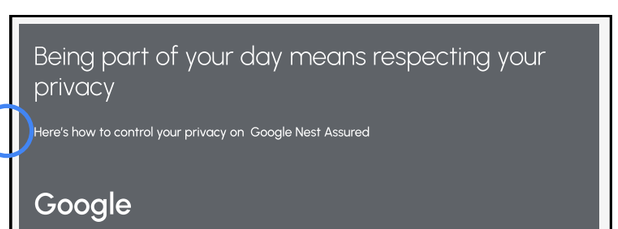
Interlocking mechanism eliminates need for adhesives or plastic and simplifies production process

Warnings

Essential warning are printed close and visible to the product

Maintaining Standard

Google is consistently committed to privacy



Ideal Packaging

Nest

Google's First Item Tracker.

Be prepared.
Stress less.

Project Plan

Month

Milestones

- Idea Validation
- Testing and Quality Assurance
- Electronic Component Coding
- Functioning Final Prototype

Updates & Revisions

- CAD Iterations
- Shape Feedback and Validation
- Coding Screen
- UX/UI Design of Screen

Learnings

- ESP32 Communication
- Mechanical Interaction
- Product Tolerances
- Environment Rendering
- Video Recording
- Video Editing

Key

- Electronics
- Video
- Administration
- CAD and Assembly

May 2024

MON	TUE	WED	THU	FRI	SAT	SUN
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	21	13	14	15	16
17	18	19	20	21	22	23

- Deciding on Product
- Component Research
- Collection of Survey Results
- Ideating
- Initial Sketching
- Prototyping Lo-Fi
- Final Decision
- User Feadback and Validation
- Choosing Brand and Identity
- Ordering Components
- Electronic Planning
- Report and Portfolio Template and Layout

June 2024

- CAD Work
- Report Writing
- Final Component Ordering
- Prototyping (3D Printing)
- Deciding Product Materials
- Coding of Screen
- User Testing
- Coding of RFID Reader
- Report Writing
- Rendering (stills)
- Assembly
- Video Renders
- Communication Between Boards
- Final 3D Prints and Manufacture
- Video Editing
- Filming Day
- Finalising Report and Slides

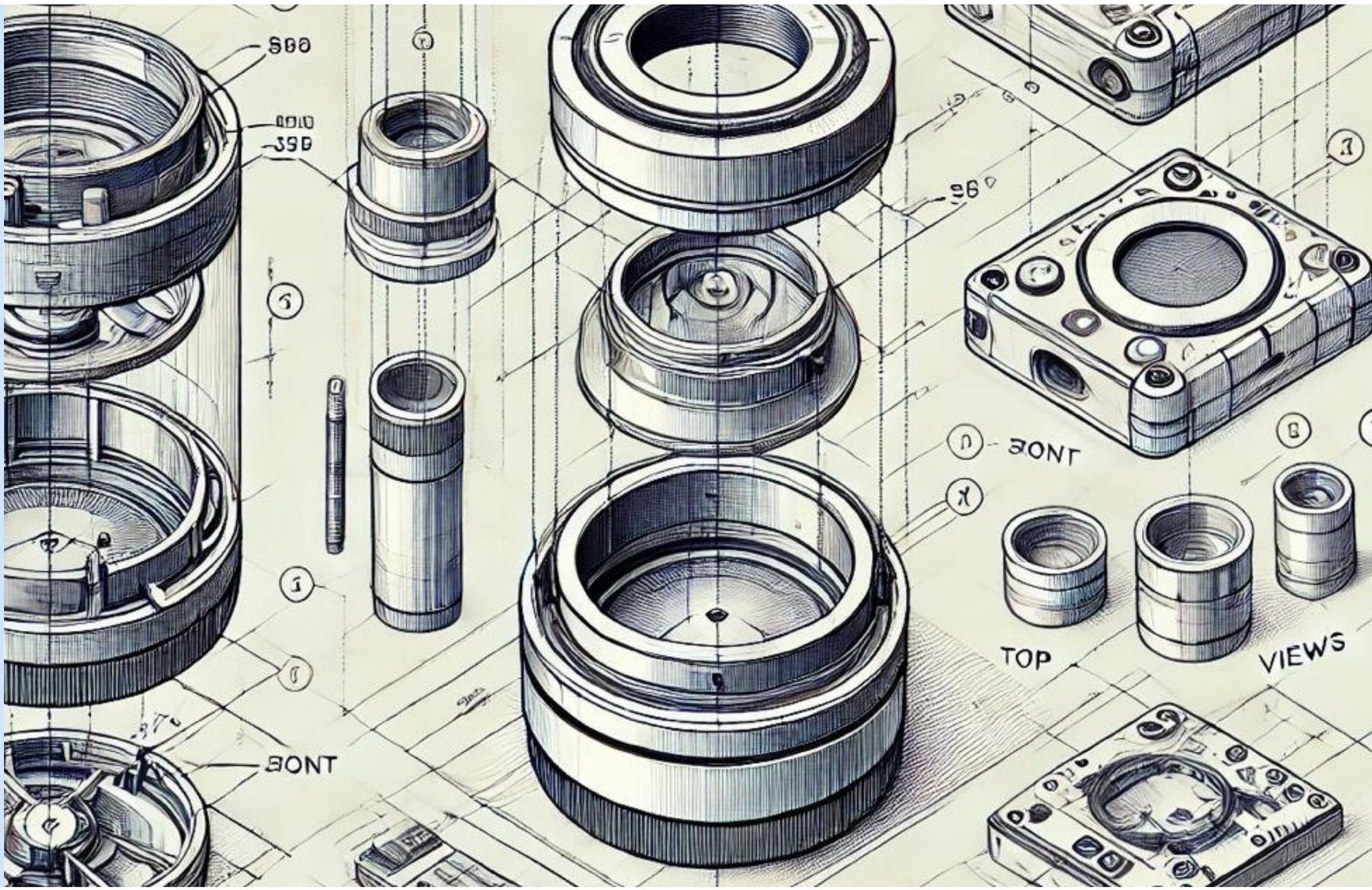
Nest Assured

RFID, ESP NOW, Mechanical Interaction

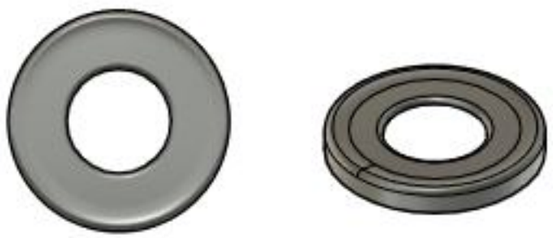
Production Data Package

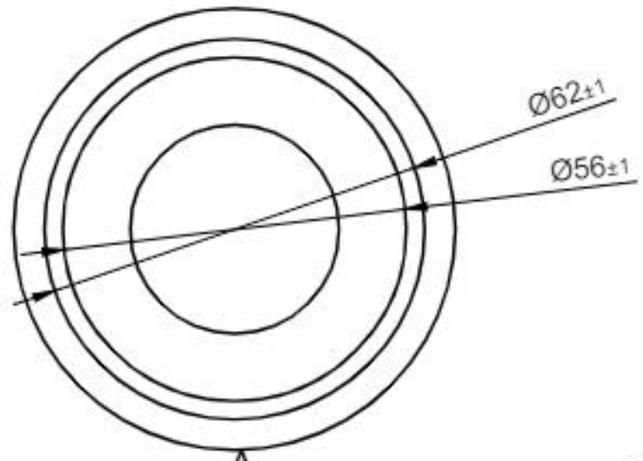
Dive inside the Nest Assured

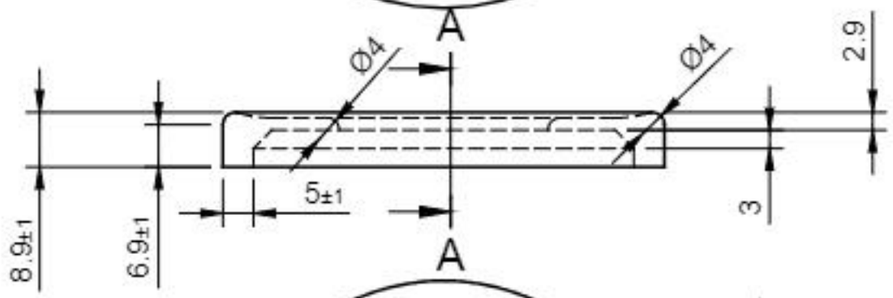
Find out more ⟶

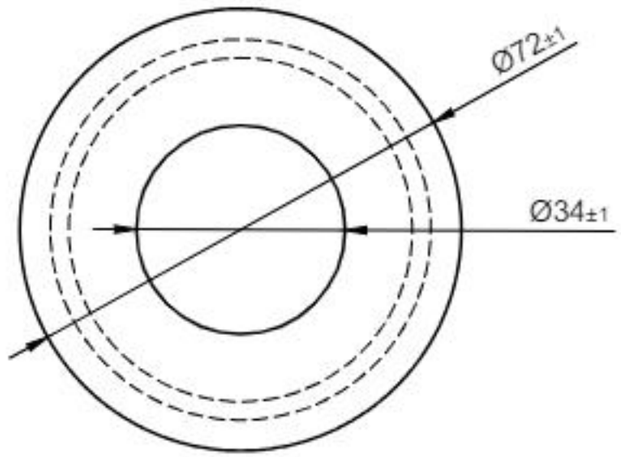


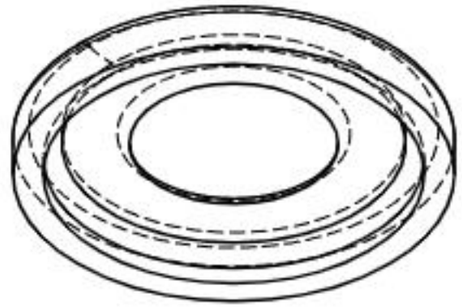
TECHNICAL DRAWING - TOP DIAL



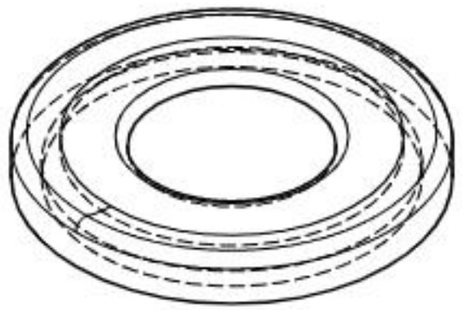








SECTION VIEW
A-A (1:1)



ALL MEASURMENTS IN MM

Material: Aluminium

MANUFACTURE METHOD: LATHE

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DE2	IDE	Josh Williams 10/06/2024	
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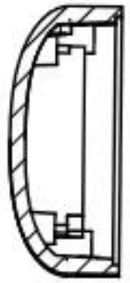
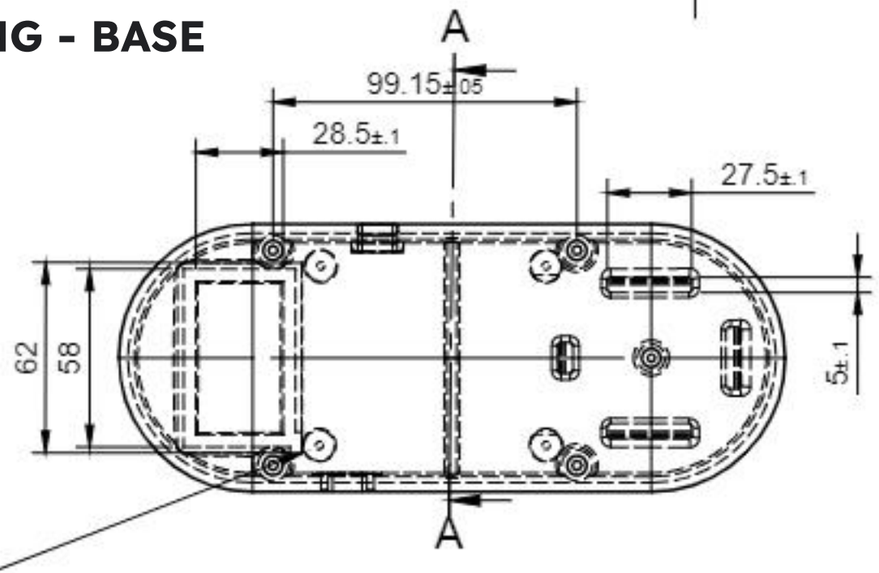
		Title: Lipped Dial	DWG No.: 0001
			Rev.: / Date of issue: 17/06/2024 / Sheet: 1/1

TECHNICAL DRAWING - BASE

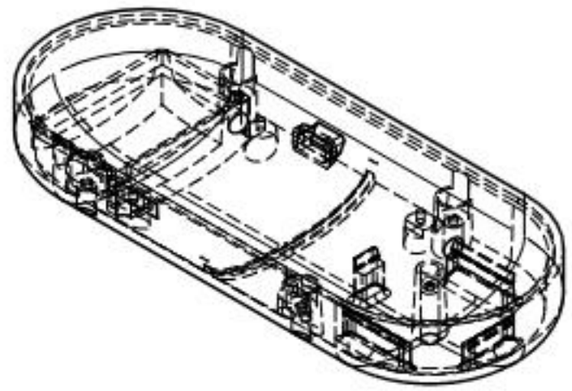
SECTION ANALYSIS

A-A (1:2)

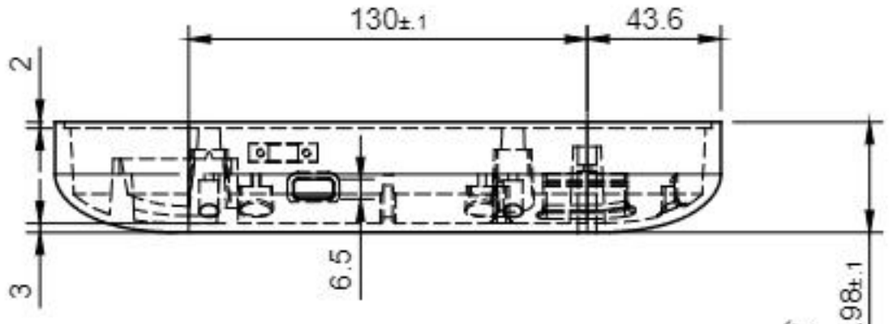


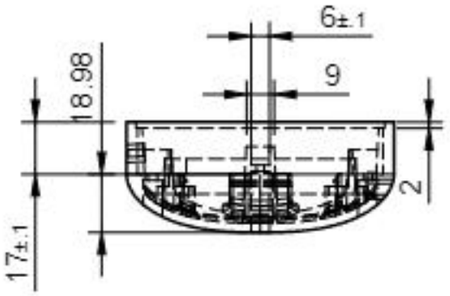


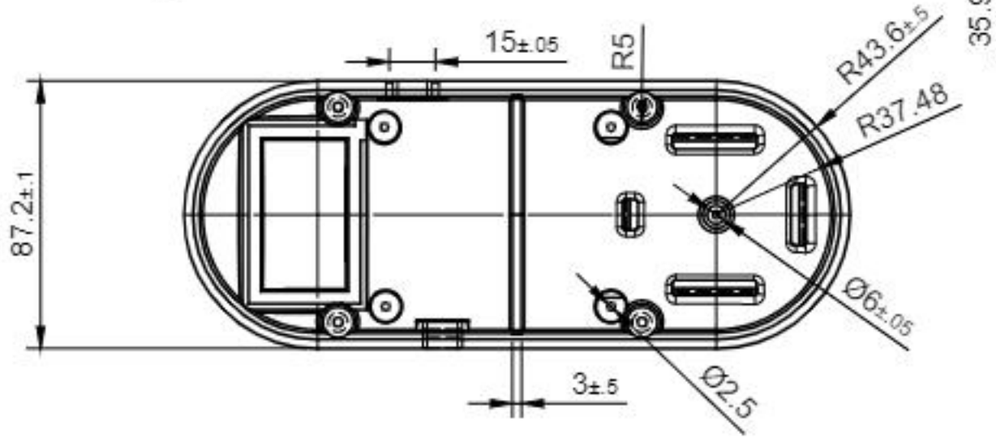
SCALE 1:3

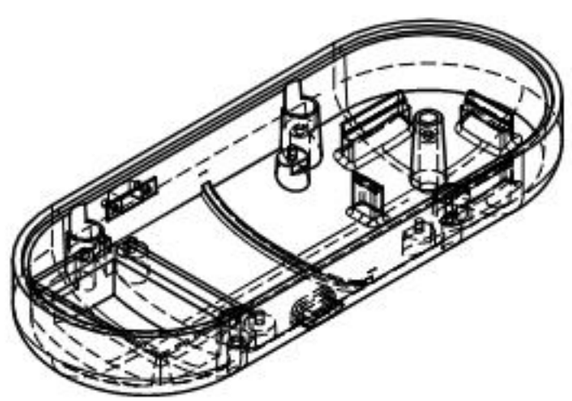


8X BOSSES MUST BE EXACT (+- 0.05MM) FOR FITMENT TO OTHER PARTS









ABS (HIGH IMPACT)

INJECTION MOULDED PART

ALL VERTICLE FACES WITH A 2 DEGREE DRAFT ANGLE

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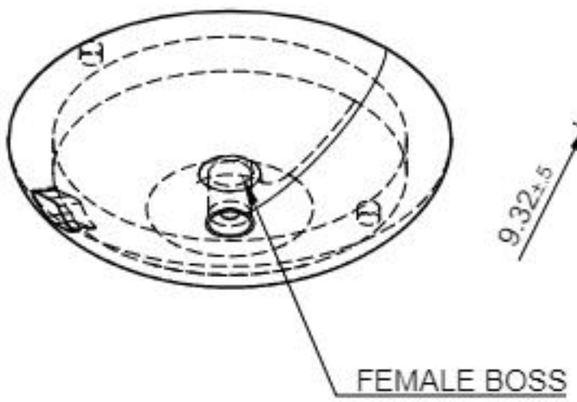
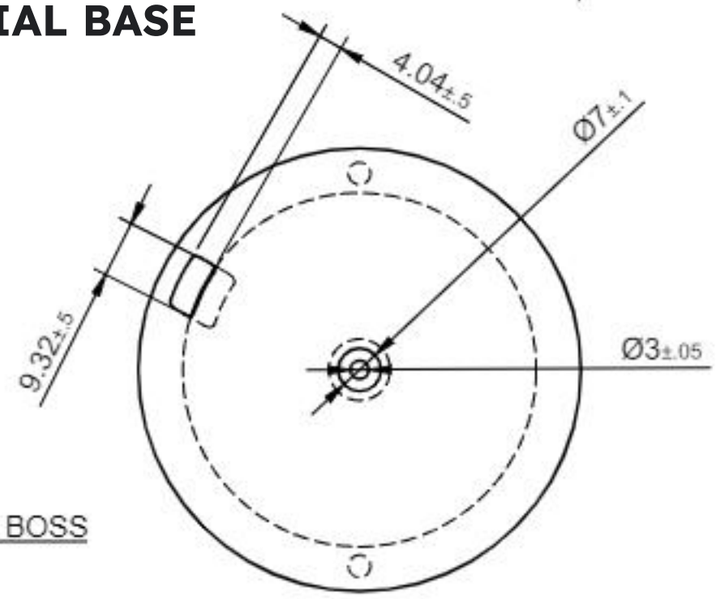
			Rev. / Date of issue / Sheet: 1/1

TECHNICAL DRAWING - DIAL BASE





FEMALE BOSS

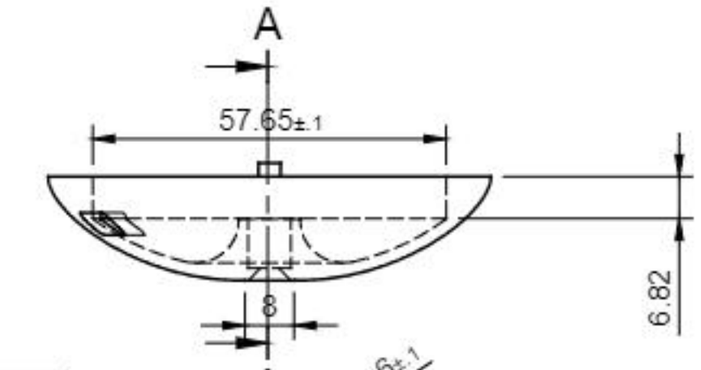
SCALE 1:1

ABS (HIGH IMPACT)

INJECTION MOULDED PART

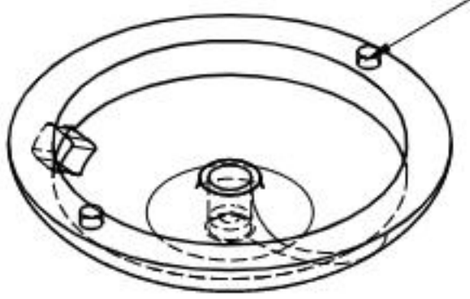
SECTION ANALYSIS

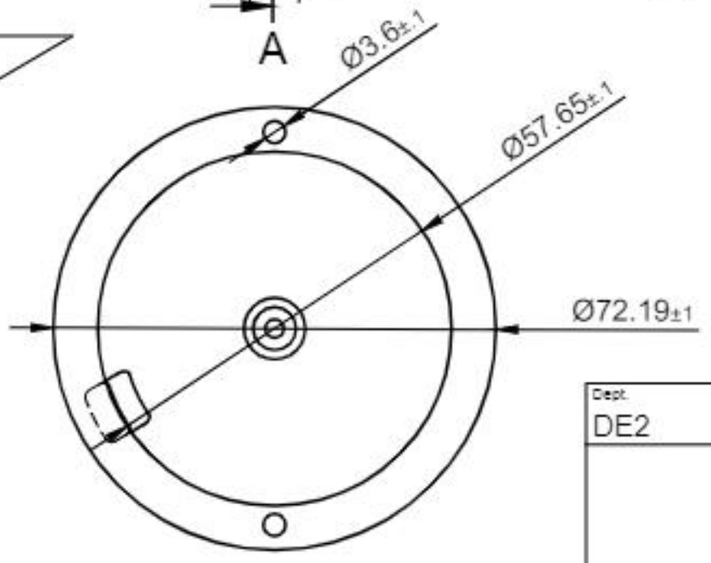
A-A (1:1)



3MM COUNTERSINK

BOSSES FOR FITMENT TO MIDDLE COMPONENT





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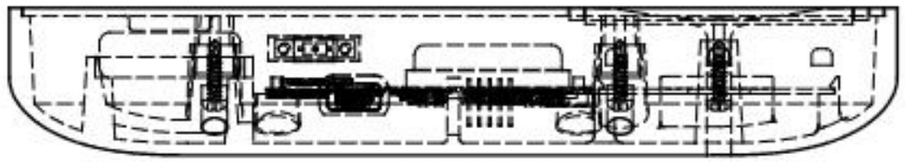
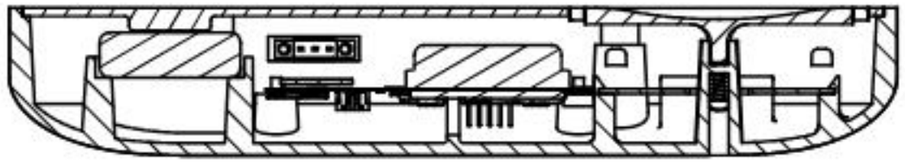
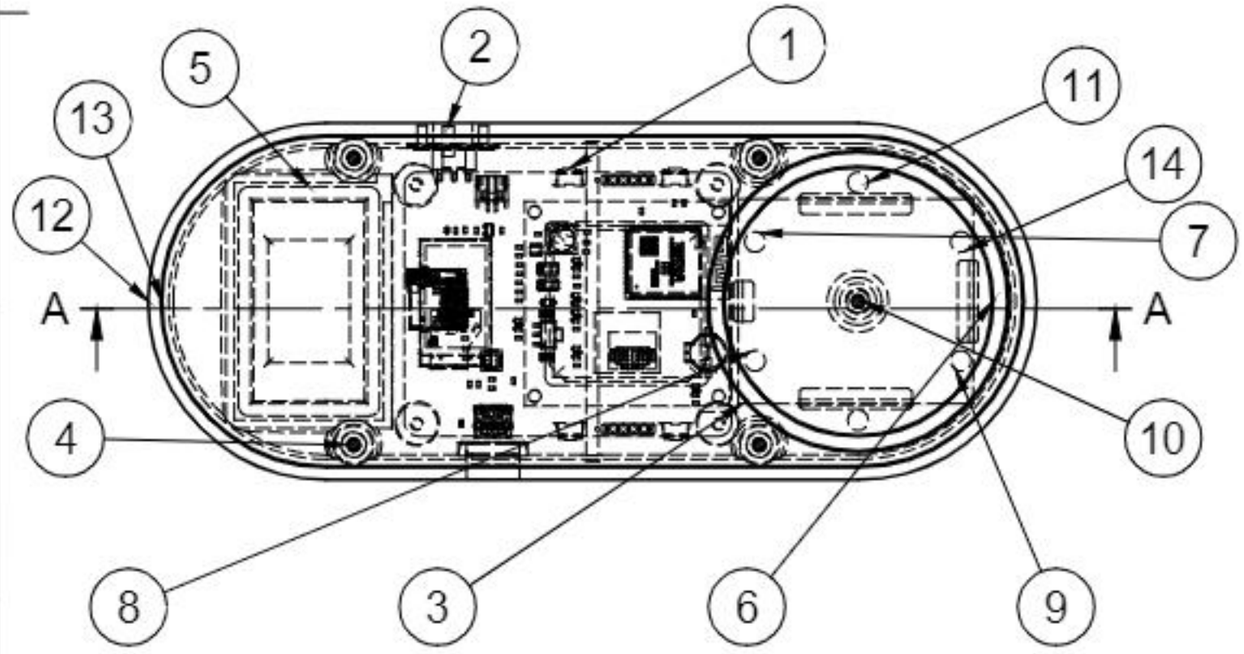
TECHNICAL DRAWING - BASEPLATE COMPONENT LIST

SCALE 1:1.5



A-A (1:1.5)
SECTION ANALYSIS





Parts List			
Item	Qty	Part Number	Material
1	1	Step UHF Reader for ESP32	STD.
2	1	Switch	STD.
3	1	LED RING PCB	STD.
4	5	Self tapping screw	STD.
5	1	42X30X9MM BATTERY	STD.
6	1	LIGHT DISK	ACRYLIC
7	1	LED	STD.
8	1	LED 2	STD.
9	1	LED 3	STD.
10	1	MIDDLE DISK	ABS
11	1	LED 4	STD.
12	1	BASE	ABS
13	1	TOP FACE	ABS
14	1	LED 5	STD.
15	1	LED 6	STD.

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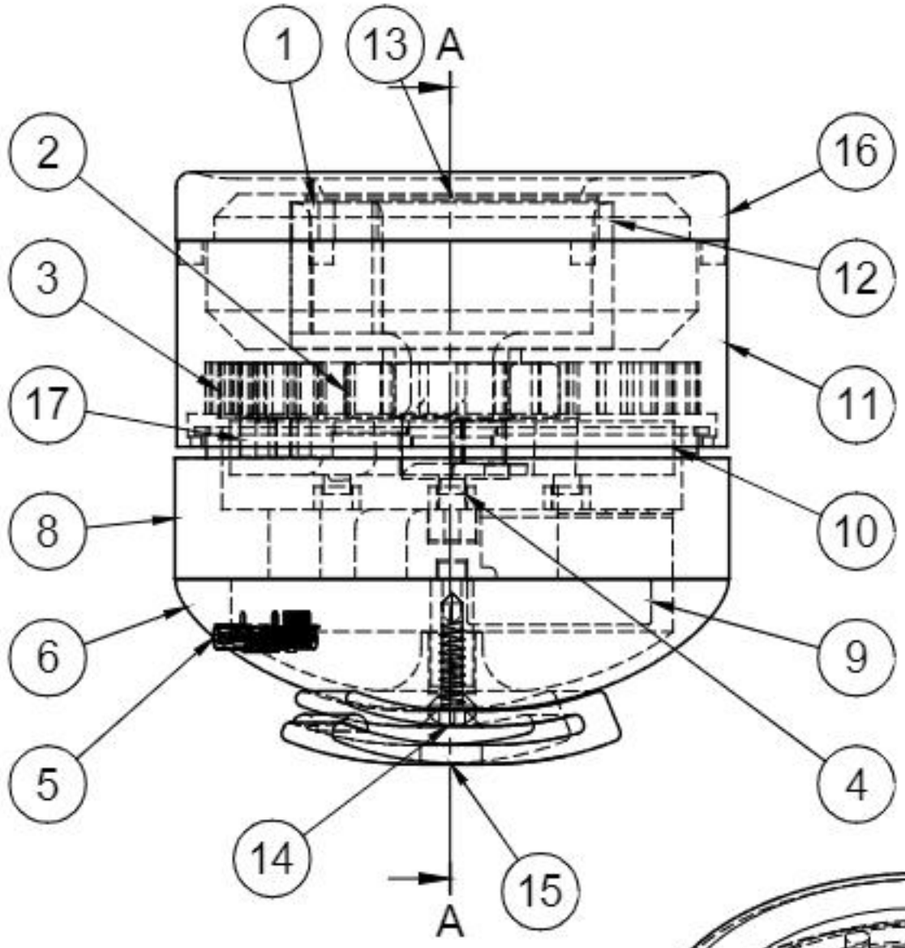
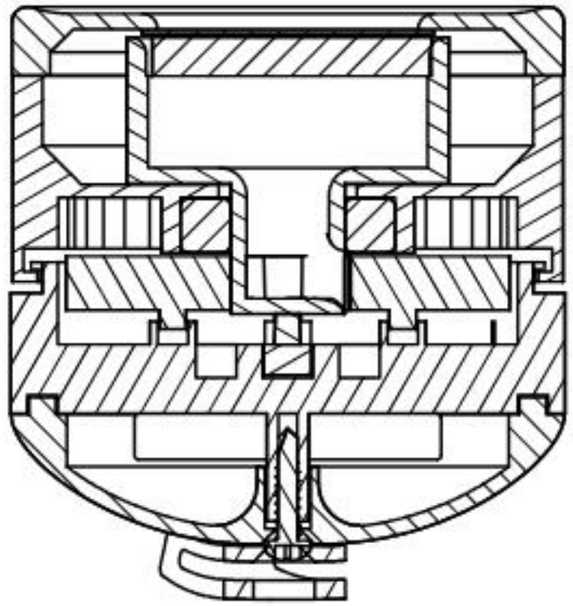
TECHNICAL DRAWING - DIAL COMPONENT LIST

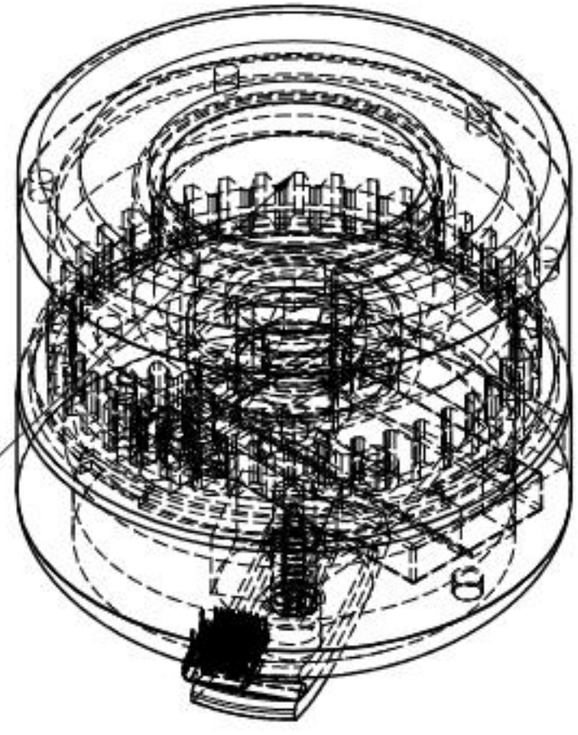


SECTION ANALYSIS

A-A (1:0.8)



SCALE 1:0.8



SPUR GEAR SETTINGS:
1.5 MODULE
PRESSURE ANGLE 20 DEGREES
GEAR THICKNES 7MM

Parts List			
Item	Qty	Part Number	Material
1	1	SCREEN	STD.
2	1	15X22X7MM BEARING	STD.
3	1	SPUR GEER (7 TEETH)	ABS
4	1	BUTTON	STD.
5	1	USB CONNECTOR TYPE C FEMALE v1	STD.
6	1	CURVED BASE 005	ABS
8	1	LOWER EXTERNAL 004	ABS
9	1	30X22X6MM BATTERY	STD.
10	1	INTERNAL DISK 003	ABS
11	1	TOP EXTERNAL 002	ABS
12	1	INTERNAL CONE 001	ABS
13	1	GLASS COVER	STD.
14	1	3X15MM SELF TAPPING SCREW	STD.
15	1	BAG CLIP	ABS
16	1	TOP DIAL	ALUMINUM
17	1	ROTARY ENCODER	STD.

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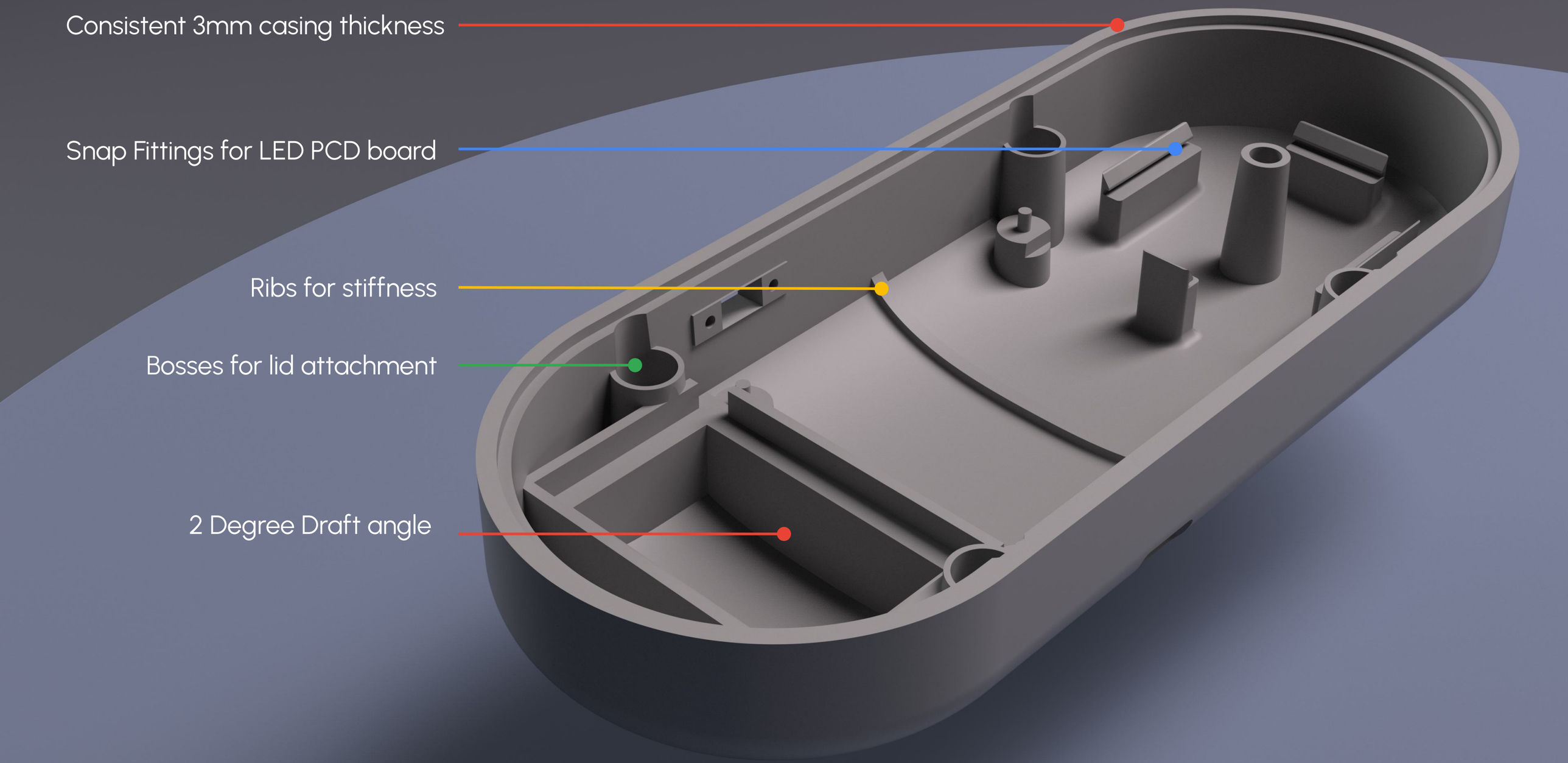
			Rev. / Date of issue: 17/06/2024 / Sheet: 1/1

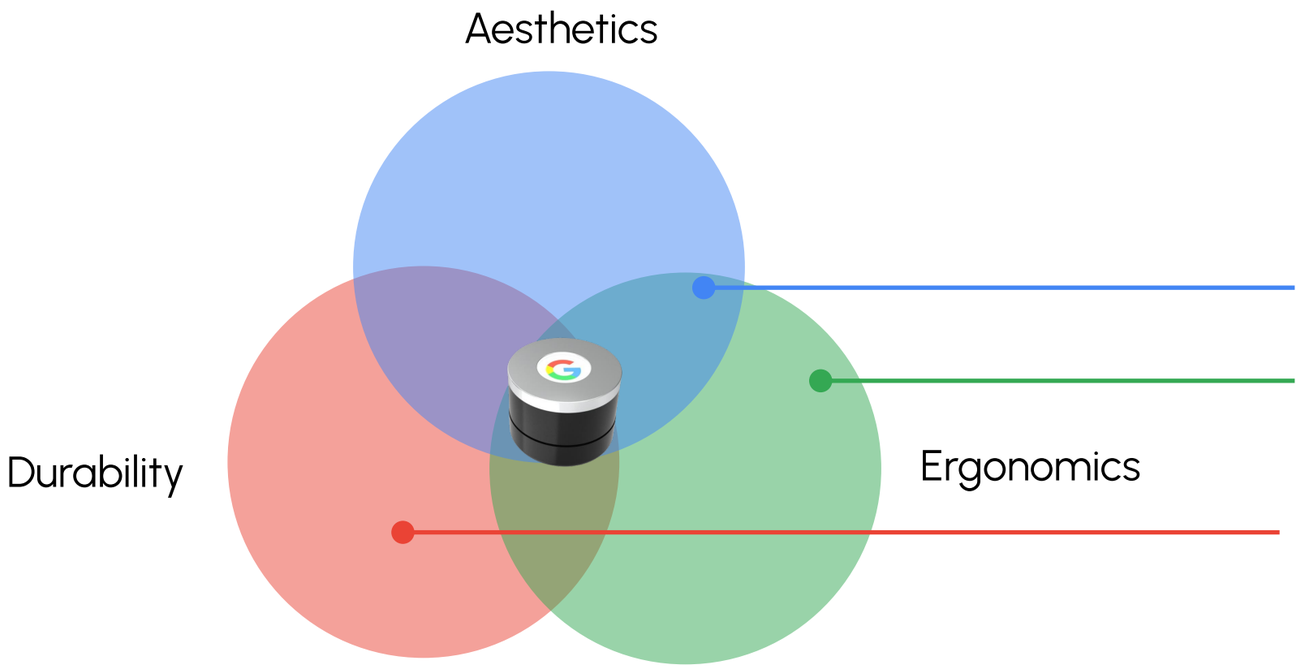
Casing Design



CMF specification

Different looks to suit your style

Nest Assured will be available in a variety of colors, in line with the google brand.



Silicone surface coating

A Black Silicone Spray coating was chosen as the finish for the Prototype. as well as a production version.

Why we chose this:

The Consistent Black finish leaves a clean and desirable look.

The grippy rubbery feel is functional to hold in the hand, and stops the device sliding around in the bottom of the users bag.

The heavy duty silicone spray can take the abuse of being thrown inside a bag without being damaged.

References

Slide 3, Facts about ADHD:

NICE UK, Attention deficit hyperactivity disorder: How common is it?, Last revised in April 2024, Available at: https://cks.nice.org.uk/topics/attention-deficit-hyperactivity-disorder/background-information/prevalence/#:~:text=In%20the%20UK%2C%20the%20prevalence,and%20in%20people%20with%20disabilities.

Slide 6 and 11, Images of Nest products:

Google Store, Your helpful Home starts here. Available at: https://store.google.com/gb/category/connected_home?pli=1&hl=en-GB

Slide 11, Google nest mission:

Nest partners, FAQ's. Available at:: https://nestpartners.withgoogle.com/faqs/#:~:text=At%20Google%20Nest%2C%20our%20mission,knows%20what%20your%20customers%20like.

Video

The video has been carefully crafted using further inspiration from Google Nest branded adverts and infographics.
Key highlights are using light, modern environments which portray the home in a functional light.

You can find the video here:

https://www.youtube.com/watch?v=iS23X2kF1yc

PRODUCT TEARDOWN

Nest

Nest Assured Technical Report

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1 Product Opportunity

1.1 Problem Statement

Young adults with Attention Deficit Hyperactivity Disorder (ADHD) struggle with object permanence, which causes them to overplan their days and schedules to future-proof themselves. They need to ensure that they keep structure to their day and have everything they need. Not having their items leads to stress and feelings of negativity.

1.2 Market Opportunity

There is a significant gap in the market for products tailored to young adults with ADHD.. Few existing solutions allow for seamless integration into daily routines. Given the fact that ADHD manifests itself differently in each person, each person will have their own unique way of managing their day and their condition. Products lacking cross-functionality and integration have continuously failed to meet the unique needs of this demographic. Additionally, existing products do not actively help to build independence and good habits, often encouraging added reliance on the product.

This presents a broader opportunity, as the challenge of forgetting items and losing structure is not unique to individuals with ADHD. People of all ages and lifestyles experience this issue. Therefore, a product designed for ADHD individuals, yet versatile enough to benefit everyone, can fill this market gap effectively.

1.3 Value Proposition

Nest Assured provides a plug-and-play tracking solution that integrates seamlessly into daily lives. By enabling cross-functionality with various brands of items and devices, and seamless integration with the Google ecosystem, Nest Assured ensures users have the peace of mind that they have everything they need for their day.

1.4 Overview

Nest Assured is a compact item scanning system which utilises Ultra-High Frequency radio frequency identification (RFID) and wireless long-range protocol ESP-NOW to ensure users have all necessary items. By facilitating effortless tracking of belongings, Nest Assured fosters healthy habits and skills, empowering users to achieve greater independence rather than reliance on the product.

With Nest Assured, you can be Rest Assured.

2 Objectives

Nest Assured is built through a user-centric design process. It features long-distance wireless communication, a seamless set-up process, and continuous improvement based on user feedback. The following quantifiable goals were outlined to ensure that the created product delivers tangible impact and empowers the users as intended.

1. **Usability and User Interface**: 90% of users must be able to successfully customise the and controls within the first use.
2. **Compatibility**: Must be compatible with at least 95% of bags in the market
3. **Set-up and Use**: 90% of users must be able to complete the setup process unaided and within their usual packing time.
4. **Security and Privacy** The product must comply with GDPR and other relevant data protection laws.

3 Concept Development Approach

3.1 Methods of Research

Research objectives were to understand the challenges faced by the user, find out what existing solutions they have used or tried, and help bridge the gap by understanding what works and what does not.

3.1.1 Secondary Research

ADHD is a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes

with human functioning or development. In the UK, approximately 4% of adults and 5% of children struggle with ADHD related symptoms. This is equivalent to approximately 1.5 million individuals (based on 2022 population statistics [1]). It is also more prevalent in males as compared to females, with a ratio of 3:1.

Socio-economically, ADHD is often linked to lower income and higher rates of unemployment. It is also more prevalent in individuals with learning difficulties. There exists 3 main types of ADHD; Hyperactive-Impulsive (ADHD-PH), Inattentive (ADHD-PI), and Combined (ADHD-C). The distribution of each type among diagnosed individuals is 15%, 30% and 55% respectively.

Currently existing solutions to ADHD fall into 5 main categories.

1. **Medication**: Takes the form of stimulant and non-stimulant medication. Help with focus and anxiety, but come with negative side-effects such as appetite suppression, sleep problems, mood swings.

2. **Therapy**: Includes one or a combination of Interpersonal Psychotherapy, Social Skills Training, and Psychological Therapies. Modifies behaviour and enhances social skills. However, therapy is not affordable for the masses. In the UK, cognitive behavioural therapy ranges from £60 to £100 per session [2].

3. **Mobile Apps**: Commonly used apps include Inflow and Tiimo. They provide a sense of community and act as a digital extension of therapy. Notifications serve as timed, frequent reminders. However, phones are distracting and most apps often requrie users to have smart watches to enable full functionality.

4. **Wearable Technology**: Provides personal, focused tracking based on biofeedback. Are always on hand and seamlessly integrates into various lfiestyles. It offers a more visible and less distracting solution compared to phones. However, they are often costly and address the symptoms rather than the root cause.

5. **Item Trackers**: Apple's AirTags and Samsung's SmartTags help locate items and notify users if items have been left behind. However, they only work for users already in the ecosystem of these brands.

3.1.2 User Research

The target demographic was young adults aged 18-25 years old. In alignment with secondary research indicating a higher prevalence of ADHD among males, special effort was made to interview females to ensure a balanced representation. Additionally, participants with each subtype of ADHD were included to capture the nuances and ensure comprehensive representation.

Interviews were chosen as the method of user research due to the varying nature of ADHD symptoms. This approach allowed participants to comfortably and fully express their thoughts, opinions, and feelings in a dynamic and intimate environment which surveys cannot replicate. An open-ended interview format was adopted, built on 3 overarching questions. Additional questions were asked based on the user's responses.

Objective	Question Asked
Understand the lifestyles and daily routines of users - what are their "must do's"?	What does your morning routine look like?
Understand what they have integrated into their daily lives to help with time management.	How do you manage your time throughout the day?
Beyond daily tasks, how do users manage other facets of their ADHD?	What do you use, if any, to manage your pain points or struggles?

Table 1: Interview Questions

3.1.3 Stakeholder Engagement

Due to the medical nature of attempting to address symptoms of ADHD, a medical

professional was also interviewed.

Astrid McGuire is a medical professional with 10 years of experience working with ADHD individuals and is certified by ADD Coach Academy, International Coaching Federation and Animas Centre for Coaching.

3 main questions were posed to Astrid:

1. What is your opinion on developing solutions and/or tools solely from the view of individuals with ADHD?
2. Would individuals with ADHD suggest solutions which would genuinely help, or might they unintentionally advance their condition?
3. What is your opinion about the techniques that individuals with ADHD develop to manage their condition on their own?

She shared essential design principles to follow, highlighted common pitfalls to avoid, and offered her perspective on how to effectively support individuals with ADHD.

3.1.4 Key Findings and Insights

Through User Research, the fundamental needs of users were identified.

1. **Impactful Products**: Products that actually resolve or help to manage the key underlying issues which give rise to the symptoms.
2. **Reward System**: Products which provide a sense of reward for completing tasks or achieving goals help contribute to motivation and continued use.
3. **Interoperability**: Each user often has their own existing system to manage their ADHD. For a product to truly deliver impact, it needs to seamlessly become part of their lifestyle and habits.

With Astrid's guidance, the four key principles on which Nest Assured is built were established. Nest Assured aims to prioritise baseline needs, build self-trust, foster independence and operate entirely as a non-subscription model

The key gap in existing solutions is aptly summarised by an interviewee who stated, "Designers have not kept in mind how to integrate it into our daily lives."

3.2 Selection Rationale

3.2.1 Initial Ideation

Based on research findings, various solutions were developed within the team.

1. **Pill Dispenser Bottle**: Retro-fitted bottle attachment with compartments for medication storage. Each compartment lights up at the specified time for medication intake.
2. **Bag Scanner**: Compact clip-on mechanism utilising RFID for item scanning during packing. Provides vibration feedback and progress lights to prevent forgetting essential items.
3. **Smart Bracelet**: Bracelet monitoring stress levels with a timer for emotional awareness. Accompanied by an app displaying well-being trends over time.
4. **Tasks Bracelet**: Community-based bracelet gamifying task completion with friends. Tasks are linked to physical buttons on the bracelet, and completing them collectively earns rewards.

3.2.2 User Feedback and Analysis

The various ideas were analysed against both quantitative and qualitative metrics, and found that Bag Scanner had the greatest development potential. A comprehensive analysis of market leaders was performed, focusing on the design, functionality, and the gaps in these products.

1. **Apple AirTags**: Offers precise tracking using Find My network. Integrates with the Apple ecosystem. However, it is limited to the Apple ecosystem
2. **Google Nest**: Enables cross-compatability with various smart devices. However, important features are tucked away in the subscription service Nest Aware [3].
3. **Tile**: A bluetooth item tracking system

which works outside of the Apple ecosystem. However, it relies on crowd-locating and may not work if there is a lack of users or if the area is less densely populated.

These observations informed the development of the Bag Scanner. The final product will consist of two main components:

1. A casing which houses the UHF RFID scanner - this is to be left at the bottom of the bag.

2. A hand-held device serving as the main user interface. It features a circular LCD screen and a navigation knob around the screen for activity selection, item viewing, and scanning activation.

RFID was chosen as it enables tagging of all items regardless of brand, without subscription-based features or reliance on crowd-locating.

4 Prototyping Plan

This section outlines the digital and physical tests that were undertaken to establish performance requirements.

4.1 Physical Tests

Component dimensions were measured to determine the necessary minimum dimensions of the product. Using Fusion360, prototypes of various shapes and input systems were designed based on the provided CAD files of the UHF Reader from the manufacturer. These prototypes were 3D printed and tested by users to ensure ease of use and intuitive user interactions.

4.2 Digital Tests

Each requirement, test procedure and results are detailed below.

Intuitive UI/UX

- Test: Prototypes of the screen were handed to users for use, and their behaviours were observed.
- Result: All users navigated the system independently without prompting or assistance. They appreciated the highlighting of selected features.

ESP-NOW Protocol Works

- Test: Protocol was set-up and packets of information were transmitted bidirectionally.
- Result: Communication protocol needed to be configured, but it works.

Wireless Communication range of at least 1.5 metres

- Test: The connection protocol was tested at distances ranging from 0 (direct contact) to 1.5 meters, increasing by 0.5 meters increments.
- Result: As per the ESP-NOW documentation, it operated within the specified range, which extends up to 220 metres.

Information sent by Scanner updates on Controller

- Test: Button styles were coded to update based on the received list of missing products.
- Result: Successful. Handlers needed to be implemented in the software to ensure that the list was received before updating the styles.

UHF RFID Reader scans tags within bag

- Test: The scanner was placed at the bottom of a bag containing the items to be scanned, as well as additional items.
- Result: Scanning range significantly increases when connected directly to a laptop for power, but decreases when connected to a battery. Electronic devices like laptops and tablets cause interference during scanning.

4.3 Physical Tests

A finite element analysis (FEA) model was set up to test whether the top lid of the bag scanner could withstand the forces it will be

subject to when inside the user's bag.

The model was set up assuming that there is a strong connection between the bosses of the top face and the base, therefore fixed supports were placed here. A fixed support was also placed around the lower edge of the lid, which replicates the lip support from the base. The force was spread equally across the entire top face of the lid.

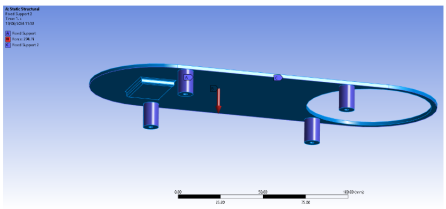

Figure 1: FEA Analysis Set Up

4.3.1 Estimating Maximum Force

The maximum force was estimated to be 20KG, this could be from having multiple heavy items placed on top of the base device such as a laptop, a full water bottle and a number of textbooks.

A 1.5 safety factor is then applied leaving a maximum force of 30KG placed on the top of the base. This equates to 294N.

4.3.2 Analysis Result

The maximum equivalent stress was found to be 20.7MPa which is below the Tensile yield strength of ABS (high impact) of 27.44MPa. Therefore the design should be able to withstand the forces of multiple items being placed on top of the device.

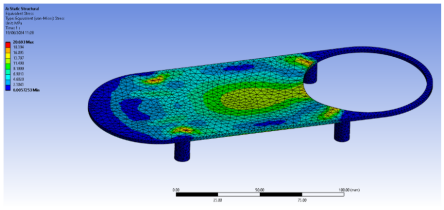

Figure 2: FEA Analysis Result

5 User Testing

Throughout the development process, users were consulted at each design iteration. The users included the six interviewees contacted during the initial user research phase. They were involved in the validation of both the physical and digital prototypes. Each physical prototype was handed to users to observe how they initially interacted with it.

As these interviewees became increasingly familiar with the ergonomics of the dial, Nest Assured was brought to street testing on the Imperial College London campus and handed to strangers for fresh insights.

5.1 Physical Testing

The following observations were made for the dial.

1. In design iterations without the aluminum dial, most users initially interacted with the sides of the body, attempting to turn it instead of using the intended dial.

2. Prior to completing the surface finish, users expressed a preference for a product with a more tactile and grip-like texture, indicating a need for enhanced user interaction.

3. Users exhibited varied usage patterns, with some operating the controller with one hand and others using two hands. No significant struggles were observed, indicating flexibility in design.

As for the scanner,

1. In the lo-fi prototyping stage, the scanner featured a flat bottom and a curved top, designed to fit into backpacks. However, user testing revealed that some users had bags with curved and slouched bottoms, while others preferred carrying the scanner in tote bags or sling bags.

2. The scanner design was modified to have a curved base and a flat top to accommodate different types of bags and user preferences.

3. Although it was not immediately intuitive for users to place the scanner at the base of the bag, it became clear after explanation. This underscores the importance of clear instructions in user guides and manuals.

Figure 3: Initial design - flat base, curved top

Figure 4: Design modification - curved base, flat top

5.2 Digital Testing

The initial UI design used rectangular buttons in the center of the screen without icons. The following feedback was received:

1. Rectangular buttons were easy to read and clear but did not make the best use of screen space.
2. Users repeatedly referenced the circular selection display of the Nest Thermostat, which aligned icons on the screen's circumference, suggesting a preference for a more efficient and intuitive design.
3. After adding icons, users suggested implementing a changing label to inform them of each icon's function. This feedback led to an updated UI design featuring circular icons with dynamic labels that convey the current process or task.

With these feedback in mind, the user interface was updated to use circular icons, featuring a label which would change and update to convey which process or task was currently running.

Figure 5: Initial UI

Figure 6: Updated UI

6 Power and Component Loads

Various different microcontrollers were considered for this project. The three best microcontrollers were found to be Raspberry Pi, Arduino, and ESP32.

The options were evaluated based on the project's use case, and the ESP32 was selected. It offered a significant advantage due to its superior power efficiency, cost-effectiveness and comprehensive documentation, and availability of diverse modules.. In addition, it supports ESP-NOW, a proprietary wireless communication protocol that allows for low-power, high-speed communication between ESP32 devices without the need for Wi-Fi connectivity. This allowed it to work wherever, regardless of WiFi availability.

6.1 Component Requirements

RFID Requirements

- Have an integrated antenna for compact design
- Able to read EU and other radio frequencies used for RFID tags around the world
- Read passive tags (stickers)
- Read multiple tags simultaneously
- Detect tags reliably to 1.5 metres of range
- Fit in a small enclosure (maximum dimensions: 200x100x50 mm)

Screen Requirements

- Be able to display colour
- Be able to display graphics (icons,

animations)

- Be compact (maximum screen diameter of 40 mm)

6.2 Used Components

6 LEDs
A light ring was to be made to convey to the user whether the device is switched on or scanning. It also adds to the engaging user experience and aesthetics. For the light to have a dynamic movement, 6 white light LED's were used to allow them to be turned on and off independently by the controller. This allows for a movement of light around the ring. A small prototype board was created with 100K Ohm resistors.

Rotary Encoder
The rotary encoder used was a 12 Pulse Incremental Mechanical Rotary Encoder with a 3.6 mm Hollow Shaft, through hole. It was selected largely due to its compact size. The dial was intended to be as slim as possible and commonly found rotary encoders would not help deliver this. Therefore an internal dial encoder was found which reduced the overall height by a factor of 15 mm and the encoder was able to sit in the perfect spot using up very little space.

Figure 7: Rotary Encoder

Button A simple tactile button with a 6mm shaft was selected to operate the click on the dial. A pull down resistor of 7.5 kOhm was placed between it and the ground.

6.3 Battery Calculations

The operating parameters of the scanner, screen and 6 LEDs are detailed below.

1. Scanner
    - **Current:** 180 mA
    - **Operating Voltage:** 3.5 V
2. Screen
    - **Current:** 1 A
    - **Operating Voltage:** 3.3 V
3. LED
    - **Current:** 30 mA
    - **Operating Voltage:** 3/5 V
    - **Power:** 105 mW

For the screen, documentation [4] recommended using a 3.7 V single-cell lithium battery with a capacity of 500mAh or below.

The LEDs in the system are versatile and can be powered by either 3 Volts or 5 Volts. To streamline the power supply, the LEDs were designed to draw power directly from the scanner. The provided electrical schematics and diagrams for the scanner, as referenced in the documentation [5], suggest the use of a 3.7 V single-cell LiPo battery. However, the documentation does not specify a particular battery capacity. Therefore, a battery with a capacity of 2000mAh was selected to ensure sufficient power availability and longer operational life.

6.4 Gear Specifications

Choosing the correct gear specs based on the purpose, material and size are very important when creating a tactile experience which will not wear down over time. As the components for the prototype were 3D printed, it was important to also choose gears which were able to be 3D printed.

The style of gear system chosen was an internal spur gear of 41 Teeth and a small external spur gear of 7 Teeth. The modulus was kept the same at 1.5. The built-in function in Fusion 360 allowed for the gears to be easily formed with the same modulus to achieve smooth operation. The 7 Tooth gear was printed with a hexagonal shaft which would go straight into the rotary encoder.

Figure 8: Gear Specifications

7 Integration Issues and Resolutions

The goal was to enable wireless communication between the scanning and controller devices using ESP-NOW, which offers an ultra-low power, long-distance, reliable bidirectional data transmission solution. Each ESP on the devices was tested to ensure they could communicate effectively while performing their respective functions. The UHF RFID reader is powered by an ESP32 S3 WROOM-1, and the LCD screen is powered by an ESP32-S3. During assembly, the dimensions of the screen, reader, and batteries were measured before purchase to ensure the overall form factor remained compact.

7.1 Software

Limited Memory

- Issue: The controller faced memory constraints as the uploaded sketch occupied 77% of the memory.
- Solution: Optimised code as best as possible, final code now occupies only 68% of memory instead.

Communication Failures

- Issue: Communication problems between the controller and scanner due to WiFi channel mismatches and mode settings.
- Solution: Troubleshot by checking and synchronising WiFi channels and ensuring correct WiFi modes.

7.2 Hardware

Connector Compatibility

- Issue: Controller that was purchased used a MX1.25 connector for LiPo power which did not match the connector which the LiPo battery came with.
- Solution: The correct connector was sourced and soldered to replace the MX1.25 connector.

Encoder Placement

- Issue: The encoder did not fit properly in its designated slot - it would come loose and interfere with the rotation.
- Solution: Secured encoder with hot glue, ensuring that it stayed in place.

3D Part Tolerances

- Issue: 3D printed parts were not fitting together correctly, likely due to tolerance issues.
- Solution: Sanded down the parts to achieve a proper fit and ensure structural integrity.

Limited Space

- Issue: Needed to fit numerous components while ensuring that the form of each part is not bulky.
- Solution: Conducted numerous design revisions to optimise space utilisation while maintaining a compact form factor.

8 Design for Manufacture

8.1 Dial

The dial on the top of the external device is made of Aluminium, largely for aesthetic

purposes as the shine of the metal stands out on top of the device. It also serves an ergonomic purpose as it highlights the dial to the user, drawing them in to touch the metal dial and explore what the device can do.

The dial was machined by hand lathing a 100mm diameter aluminium rod into the desired shape. 45°chamfers were placed on the topmost and inside edges. The industrial manufacture process for this would not differ greatly, and would be possible with a CNC lathe. It is also preferable to achieve a brushed finish rather than a polished one.

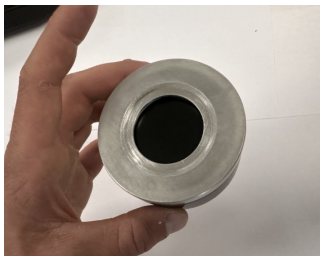

Figure 9: Aluminium Dial

8.2 Casings

The wall and rib thickness play a crucial role in achieving the right balance between stiffness, strength, and weight for the component casing. Several parts within the casing will be manufactured via injection moulding, with the base of the scanner being the most complex and prominent piece. Therefore, uniform wall thickness is essential to reduce the risk of defects [6].

According to standard design guidelines, wall thickness for reinforced materials typically ranges from 0.75mm to 3mm. Given that the device will experience impact and weight from items placed on top of it, a thickness of 3mm is chosen to ensure adequate support for the casing, using ribs and supports as needed.

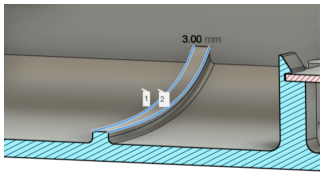

Figure 10: Wall Thickness

Additionally, 2mm fillets were chosen for the corners of ribs and edges to further facilitate mould release and reduce stress concentrations.

8.2.1 Designing for Injection Moulding

To ensure easy release from the mould, injection moulded parts require specific design considerations. These include avoiding overhangs in the direction of mould release and applying a draft angle of 1.5-2 degrees on vertical surfaces. Smoothing out sharp corners with a radius of at least 0.5 times the wall thickness is also essential to reduce stress concentrators and aid in mould release.

In adherence to these guidelines, a 2-degree draft angle has been implemented on all vertical surfaces within the casing.

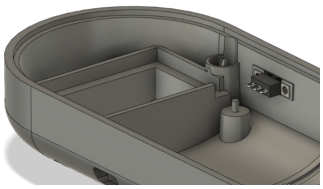

Figure 11: Vertical Surface

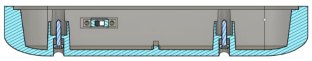

Figure 12: Draft Angles

8.2.2 Self-Tapping Screw Bosses

To ensure a secure attachment between the top surface and the bottom casing. The top face secures the RFID reader and battery while also enhancing structural rigidity. 4 self-tapping screws were used with male-to-female bosses. The female side is mounted on the underside of the case for additional rigidity. Self-tapping screws were chosen as they are useful when disassembling and reassembling products since they will insert themselves into the same threads they have been removed from [7]. The chosen screws have a head diameter of 6mm and a length of 15mm to maintain a compact size while ensuring sufficient thread engagement.

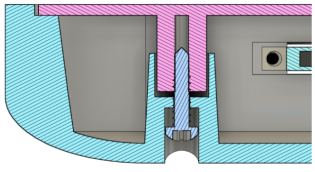

Figure 13: Self-Tapping Screw Bosses

9 Branding Considerations

Branding enhances the market position of a product. It helps users associate a product's visual identity, values, and quality with a well-established brand. This is particularly important when addressing specific challenges, such as ADHD-related forgetfulness.

Google Nest was chosen for its established reputation for innovation, reliability, and user-centric design. Their core mission [8] revolves around empowering users to stay informed, feel secure, and monitor their surroundings effortlessly. This aligned with the project goals and made Google Nest the ideal choice.

By integrating the Google branding, the products aims to embody the following key values that are integral to the brand's identity:

- **User-centricity**: Strong focus on user experience, ensuring they are intuitive, accessible, and useful.
- **Innovation**: Integrate cutting-edge technology.
- **Simplicity**: Ensure the product is intuitive while ensuring a clean, minimal design.

9.1 Brand Identity Alignment

Google's mission is to "organise the world's information and make it universally accessible and useful"[9]. Nest Assured aims to streamline users' daily organisation by quickly identifying and displaying essential items needed throughout their day. With the potential to integrate seamlessly into the Google ecosystem, it offers a smart and intuitive item-tracking experience that complements users' lifestyles. This alignment with Google's vision facilitates future developments, such as automatically syncing with calendars to suggest needed items based on scheduled events.

9.2 Naming Significance

The name "Nest Assured" aligns with Nest's branding strategy of conveying security, reliability, and peace of mind. "Nest" reflects the brand's focus on creating a comfortable, connected home environment, while "Assured" emphasises the product's role in ensuring users have all the items they need, providing confidence and certainty. It suggests trust and dependability, reinforcing the brand's commitment to simplifying and enhancing daily lives.

9.3 Target Audience

Google's vast user base provides a broad target audience. Key demographics include:

- **Students and Professionals**: Those who lead dynamic and active lifestyles and need to manage their belongings efficiently.
- **Frequent Travellers**: Users who rely on Google Maps, Google Travel, and other services for their travel needs and can benefit from enhanced item-tracking capabilities.
- **Google Users**: Individuals already engaged with Google products and services, who value seamless integration and consistency across their devices.
- **Tech Enthusiasts**: Early adopters who are keen on the latest technological advancements and appreciate innovative solutions.

9.4 Visual Identity

Nest Assured was developed in alignment with Google's branding guidelines detailed in their Partner Marketing Hub and Brand Resource Centre [10]. Throughout the development process, Google's brand element guides were adhered to.

9.4.1 Colour Scheme

Google's colour scheme was obtained from their Partner Marketing Hub [11]. Their webpages were also studied to understand how the different colours were used to highlight and display content.

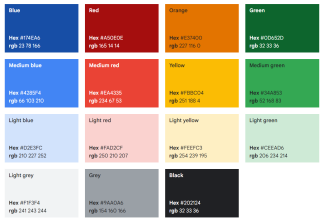

Figure 14: Google's Colour Scheme [11]

9.4.2 Typography

Google uses Product Sans and Google Sans as its official fonts. These fonts are proprietary and exclusively available for use within Google products. They are not licensed for external use, as detailed in Google's official guidelines [12].

To adhere to these guidelines while maintaining visual alignment, Urbanist was selected as an alternative font. It closely mirrors the profile of Google Sans, ensuring visual coherence with the Google Nest ecosystem. This choice respects Google's font usage policies while achieving a consistent and recognisable design aesthetic.

10 Design for Assembly

In the design of the casings for both the controller and the scanner, several features were incorporated to facilitate product assembly.

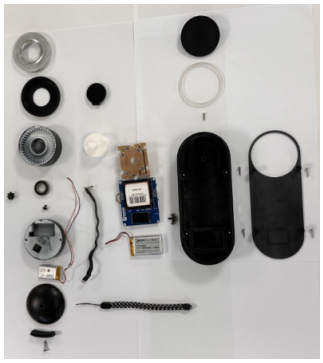

Figure 15: Product Teardown

These included alignment aids such as ribs, grooves, and raised edges to ensure proper positioning of components like the reader and screen. Additionally, alignment marks and precisely positioned screw holes were added for accurate mounting of components. Snap-fit mechanisms were also integrated to securely hold the PCB in place.

Figure 16: Sanding down of snap-fits

10.1 Product Assembly Process

The entire process was timed and took 14 minutes and 29 seconds. Based on an assembly cost estimate of 7 USD per hour, this totalled out to USD$1.70.

10.1.1 Dial

Key steps include:

1. Press bearing into middle body
2. Place rotary encoder ronto holder
3. Attach button to underside of lower body
4. Attach battery and feed cable through
5. Attach base
6. Screw clip-on onto the base
7. Attach screen to the holder
8. Attached lathed top

Figure 17: Dial Assembly Process

10.1.2 Scanner

Key steps include:

1. Put charging plate through ring
2. Place both parts onto top cover
3. Lay UHF RFID Reader and push PCB into snap fits
4. Attach battery and connect
5. Feed power button through cut-out slot
6. Add cotton gauze for light diffusion

7. Place lid
8. Screw close the case

Figure 18: Scanner Assembly Process

11 Compliance Research and Requirements

To ensure legal compliance and customer confidence in the product's quality and safety, current compliance requirements were adhered to during product development. As an international company, Google takes sole responsibility for ensuring their products conform to all relevant statutory requirements. Adhering to EU and UK standards is particularly crucial due to these regions being a significant portion of their market.

11.1 European Union

11.1.1 CE Marking

The CE marking [13] is a mandatory conformity mark for products sold within the European Economic Area, which ensures that the product meets EU requirements. To achieve the CE marking, Google must:

- Ensure product complies with relevant EU directives
- Conduct a conformity assessment in house or by a notified body
- Compile comprehensive technical documentation
- Draft and sign the Declaration of Conformity
- Affix CE marking to the product visibly, legibly and indelibly
- Maintain compliance and post-market surveillance

11.1.2 New Legislative Framework

Since 2008 a new legislative framework [14] has been adopted to improve the internal market for goods and strengthen the conditions for placing a wide range of products on the EU market. The following legislations apply:

1. **Radio Equipment Directive(RED) 2014/53/EU**
    - Ensures that radio equipment placed on the market meets essential requirements for health, safety and efficient use of radio spectrum. Nest Assured needs to abide to this as it employs wireless communication.
2. **Electromagnetic Compatibility Directive (EMC) 2014/30/EU**
    - Ensures that electronic devices do not emit excessive electromagnetic interference and are not adversely affected by it.
3. **Low Voltage Directive (LVD) 2014/35/EU**
    - Ensures that electrical equipment operates safely within specified voltage limits to protect users from electrical hazards
4. **Restriction of Hazardous Substances Directive (RoHS) 2011/65/EU**
    - Limits the use of specific hazardous materials in electrical and electronic equipment to reduce environmental and health risks.
5. **Eco-design Directive 2009/125/EC**
    - Establishes requirements for the energy efficiency and environmental impact of energy-related products throughout their life cycle.
6. **Waste Electrical and Electronic Equipment Directive (WEEE) 2012/19/EU**
    - Addresses the treatment, recycling, and disposal of electronic waste to promote environmentally sound management.
7. **General Product Safety Directive (GPSD) 2001/95/EC**

- Ensures that products placed on the market are safe for consumers, covering aspects not addressed by specific sectoral legislation.

8. **Battery Directive 2006/66/EC**

    - Regulates the manufacturing, disposal, and recycling of batteries to minimize environmental impact and promote sustainability.

11.2 Incorporating Industry Standards

Google adheres to key industry standards to ensure the highest quality and safety of its products. These designated standards ensure that products complying with them are deemed to be safe in relation to the areas they cover.

1. **Radio Equipment Directive(RED) 2014/53/EU [15]**

    - EN 50566 (for hand-held and body mounted device in close proximity to the human body)
    - EN 300 328 (for wideband transmission systems)
    - EN 300 440 (for short range devices)
    - EN 300 720 (for Ultra-high frequency on-board vessels communications systems and equipment)

2. **Electromagnetic Compatibility Directive (EMC) 2014/30/EU [16]**

    - EN 55032 (Electromagnetic compatibility of multimedia equipment- emission requirements)

3. **Low Voltage Directive (LVD) 2014/35/EU [17]**

    - EN 60950 (Information technology equipment - safety and general requirements)
    - EN 62368 (Audio/video information and communication technology equipment-Safety requirements)

4. **Restriction of Hazardous Substances Directive (RoHS) 2011/65/EU [18]**

    - EN 50581 (Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances)

5. **Eco-design Directive 2009/125/EC**

    - EN 50564 (Electrical and electronic household and office equipment-Measurement of low power consumption)
    - EN 60960-1 (Requirements for the reduction of hazardous substances in multimedia equipment)
    - EN 62087-1 (Audio, video, and related equipment – Methods of measurement for power consumption)

6. **Waste Electrical and Electronic Equipment Directive (WEEE) 2012/19/EU**

    - EN 50625 (Collection, logistics and treatment requirements for WEEE)

7. **General Product Safety Directive (GPSD) 2001/95/EC**

    - EN 60335-1(Household and similar electrical appliances - Safety)
    - ISO 10377 (Consumer product safety - Guidelines for suppliers)

8. **Battery Directive 2006/66/EC**

    - EN 62133 (Secondary cells and batteries containing alkaline or other non-acid electrolytes- Safety requirements for portable sealed secondary cells, and for batteries made from them, for use in portable applications)
    - IEC 60086-4 (Primary batteries-Safety of lithium batteries)

11.3 UK Conformity Assessed (UKCA) Marking

Post-Brexit, the UKCA marking has been introduced as an accepted marking or combination of markings for the CE marking for manufactured products placed in the Great Britain market. This indicates that the product complies with regulatory requirements tailored to the UK. To attain the UKCA marking, Google must:

- Ensure product complies with applicable UK regulations and standards
- Perform a conformity assessment in house or by a notified body
- Compile comprehensive technical documentation
- Draft and sign the Declaration of Conformity
- Affix UKCA marking to the product visibly, legibly and indelibly
- Maintain compliance and post-market surveillance

12 Labels and User Guides

12.1 Labels

12.1.1 Branding

The Google logo and “Nest Assured” product name are displayed in the brand colours and theme on the packaging to convey a strong brand identity. Additionally, the product code and serial number are included for easy identification, imperative for customer support and warranty claims.

12.1.2 Compliance Markings

The user guide will be marked with the following compliance markings.

- CE Mark: Indicating conformity with health, safety, and environmental protection standards for products sold within the European Economic Area (EEA).
- FSC Logo: If the packaging uses paper or cardboard, the Forest Stewardship Council (FSC) logo indicates that the materials come from responsibly managed forests.
- Green Dot: Shows that a financial contribution has been made to a qualified national packaging recovery organisation, supporting the recycling of packaging materials.

12.2 User Guides

To provide a seamless experience for our users, a comprehensive and user-friendly guide was developed. Our product offers numerous features that users need to be effectively informed about.

The guide must offer:

- **Clarity:** provide clear, simple, concise instructions on the setup and comprehensive coverage of all the product’s features and functions.
- **Accessibility:** Ensure the guide is accessible and easily understandable.
- **Troubleshooting:** Provide common issues and solutions, along with maintenance tips.
- **Safety Information:** Present safety warnings, usage precautions and maintenance care.
- **Support:** Offer contact information and additional resources.

A physical manual was included within the packaging. This concise guide provides step-by-step instructions for setting up the device smoothly. An additional online user guide contains more detailed information, conveyed through interactive elements like video. To ensure all users could easily find all the necessary information, a QR code and web link were provided inside the manual.

12.3 Safety Information

Inside the packaging, a manual with important safety information is provided to ensure the correct use and safe handling of the

product. It highlights essential aspects, including environmental conditions that should be avoided as it might damage the product, as well as disposal guidelines.

Figure 19: Packaging Labels with Concept Image

12.4 Specification of Materials

The manuals will be made of FSC certified recycled paper. Documentation band wraps will be obtained from Google's listed supplier [19], Arjowiggins, with a grade of GT-63-HM. Arjowiggins is an expert in FSC-certified pulp [20].

12.5 Production Processes

The production process for paper will be through pulping and forming of wood pulp.

13 Product Packaging

13.1 Branding and Visual Identity

The packaging reflects Google's high-end and innovative brand identity as the first user touchpoint. The design is simple, clean, and minimalist, featuring a white color scheme consistent with Google's existing packaging. The Google logo and product name are prominently displayed, with a consistent color scheme and a high-quality image of the product on the back.

13.2 Target Audience

Google's packaging targets a diverse audience, including tech enthusiasts, professionals, and everyday consumers seeking smart home solutions or mobile devices. The design is accessible and attractive, conveying sophistication and reliability. Its clean and straightforward aesthetic appeals to various age groups and tech knowledge levels, making the products approachable and desirable.

13.3 Functionality and Design

Google emphasises user experience, ensuring packaging is easy to open with clear, concise setup instructions. The packaging utilises a mechanical seal in the form of a perforated strip design. Peeling off the strip allows for easy unboxing.

Internal compartments are well-organised, securely holding all components for easy access. The unboxing experience is intuitive and enjoyable.

13.4 Material Selection

13.4.1 Physical Product

The physical product features an aluminium dial and 3D-printed components with a soft-touch rubber surface finish that mimics the feel and appearance of leather.

13.4.2 Packaging

Material selection for product packaging is guided by principles of sustainability and durability. Google outlined its objective of achieving "Plastic-free packaging by 2025" [21] and has broader goals to transition to a circular economy [22], ensuring that all product packaging is completely plastic-free and recyclable. In line with these principles, Google's plastic-free packaging design guide and material library [19] were referenced.

Cardboard was chosen for the panels due to its stiffness and ease of recyclability in the UK. Specifically, for the folding carton retail box, 300 gsm paper board will be used. Product wrapping will be sourced from SIP, using grade SIP-11 material, and box closure labels will be supplied by UPM.

Instead of the currently used thermoformed PET trays which hold and protect the product, corrugated cardboard would be used. Although softer than PET, it would still offer comparable protection while increasing recyclability and providing a natural aesthetic.

Following the plastic-free packaging design guide, this would specifically be 250 gsm top sheet corrugated cardboard.

These materials will protect the product during shipping and handling while also being highly recyclable. Additionally, non-toxic inks and adhesives are used to enhance the recyclability of the packaging. This comprehensive approach underscores Google's dedication to sustainability and environmental responsibility, ensuring high-quality, durable, and recyclable packaging.

13.4.3 Compliance Markings

Necessary labels were added to the external packaging to comply with UK regulation, including adherence to guidelines for environmental impact, safety, and labelling.

- **CE Mark**: Indicating conformity with health, safety, and environmental protection standards for products sold within the European Economic Area (EEA).
- **WEEE Symbol**: Indicating that the product should not be disposed of with regular household waste and must be recycled according to the Waste Electrical and Electronic Equipment (WEEE) Directive.
- **Mobius Loop**: Indicating that the packaging is recyclable.
- **FSC Logo**: If the packaging uses paper or cardboard, the Forest Stewardship Council (FSC) logo indicates that the materials come from responsibly managed forests.
- **RoHS Compliance**: Indicates that the product complies with the Restriction of Hazardous Substances Directive, limiting the use of specific hazardous materials found in electrical and electronic products.
- **Green Dot**: Shows that a financial contribution has been made to a qualified national packaging recovery organization, supporting the recycling of packaging materials.
- **Tidy Man Symbol**: Encourages consumers to dispose of the packaging properly in a waste bin.

14 Final Product

The video can be accessed via this link, or through https://youtu.be/iS23X2kF1yc.

14.1 Product

The final product is designed as such.

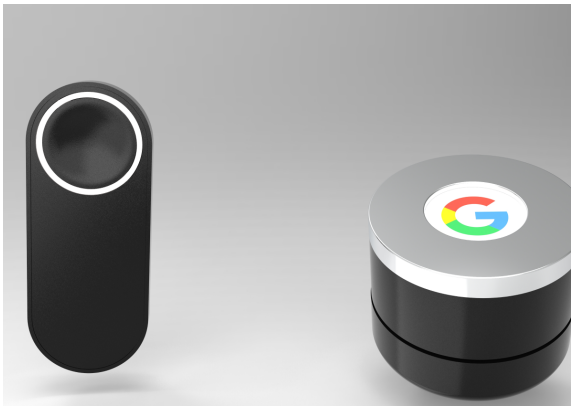

Figure 21: Final Product

14.2 Packaging

The final packaging is designed as such.

Figure 22: Packaging

14.3 Concept Images

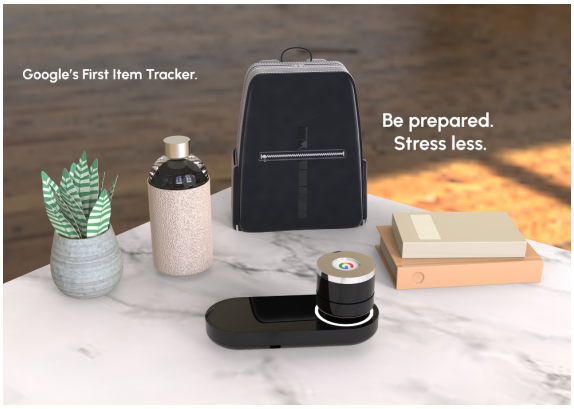

Figure 23: Concept Image 1

15 Conclusion

Over the past two terms, extensive user research, testing, and consultations with a medical professional were conducted during the initial idea development stages. Based on the insights gathered, a double diamond strategy was adopted. This involved the team diverging to ideate four different solutions, each addressing unique problems. These ideas were then evaluated using both quantitative and qualitative metrics, resulting in the selection of the most promising concept.

The physical development phase included an exploration of various forms, ergonomics, and structures for each component, ultimately leading to an optimized product design. Branding considerations were strictly adhered to, and compliance measures were diligently followed to ensure the product met all regulatory requirements. Additionally, materials were chosen not only to align with company policies but also to minimize environmental impact and provide adequate protection for the product.

Rigorous digital and physical testing confirmed that the product performs as expected. This comprehensive approach ensured that the final product is not only functional and compliant but also environmentally responsible and well-branded.

References

[1] O. for National Statistics, "Estimates of the population for the uk, england, wales, scotland, and northern ireland." [Online]. Available: https://www.ons.gov.uk/peoplepopulationandcommunity/populationandmigration/populationestimates/datasets/populationestimatesforukenglandandwalesscotlandandnorthernireland

[2] N. H. S. (NHS), "Overview - cognitive behavioural therapy (cbt)." [Online]. Available: https://www.nhs.uk/mental-health/talking-therapies-medicine-treatments/talking-therapies-and-counselling/cognitive-behavioural-therapy-cbt/overview/

[3] Google, "Nest aware." [Online]. Available: https://store.google.com/gb/product/nest_aware?hl=en-GB

[4] waveshare, "Esp32-s3-lcd-1.28." [Online]. Available: https://www.waveshare.com/wiki/ESP32-S3-LCD-1.28#Hardware_Description

[5] sb components, "Uhf reader esp32 sch." [Online]. Available: https://github.com/sbcshop/UHF_Reader_ESP-32_Hardware/blob/main/Design%20Data/UHF%20Reader%20ESP32%20SCH.pdf

[6] U. P. M. (UPM), "Why is uniform wall thickness important?" [Online]. Available: https://www.upminc.com/resource/why-is-uniform-wall-thickness-important/#:~:text=Uniform%20wall%20thickness%20is%20a,and%20non%2Dcritical%20design%20failures

[7] R. Components, "Self-tapping screws guide." [Online]. Available: https://uk.rs-online.com/web/content/discovery/ideas-and-advice/self-tapping-screws-guide

[8] Google, "Introducing google nest." [Online]. Available: https://partnermarketinghub.withgoogle.com/brands/google-nest/overview/overview/

[9] ——, "About." [Online]. Available: https://about.google/intl/ALL_uk/

[10] ——, "Brand element guidelines." [Online]. Available: https://about.google/brand-resource-center/brand-elements/

[11] ——, "Color palette." [Online]. Available: https://partnermarketinghub.withgoogle.com/brands/google-news/visual-identity/color-palette/

[12] G. Fonts, "Frequently asked questions." [Online]. Available: https://developers.google.com/fonts/faq#:~:text=Google%20Sans%20is%20not%20available,exclusively%20used%20in%20Google%20products

[13] Google, "Declaration of conformity for nest devices." [Online]. Available: https://support.google.com/product-documentation/answer/7477422?hl=en#zippy=%2Cgoogle-nest-mini-nd-gen%2Cuk

[14] E. Parliament and of the Council, "Directive 2014/53/eu of the european parliament and of the council," *Official Journal of the European Union*.

[15] D. F. BUSINESS and TRADE, "References to standards for radio equipment in support of the radio equipment regulations 2017 (s.i. 2017/1206)." [Online]. Available: https://assets.publishing.service.gov.uk/media/65bb774c4965c5000de8a344/ds-0095-24-radio-equipment-notice.pdf

[16] E. . I. S. DEPARTMENT FOR BUSINESS, "References to standards for electromagnetic compatibility in support of the electromagnetic compatibility regulations 2016 (s.i. 2016/1091))." [Online]. Available: https://assets.publishing.service.gov.uk/media/6356a

7ced3bf7f0bd6f51773/ds-0071-22-emc-notice.pdf

[17] D. F. BUSINESS and TRADE, “References to standards for electrical equipment designed for use within certain voltage limits in support of the electrical equipment (safety) regulations 2016 (s.i. 2016/1101).” [Online]. Available: https://assets.publishing.service.gov.uk/media/6643285bae748c43d3793a27/ds-0097-24-low-voltage-equipment-notice.pdf

[18] F. DEPARTMENT FOR ENVIRONMENT and R. AFFAIRS, “References to standards for the technical documentation required for assessing materials, components and electrical and electronic equipment in support of the restriction of the use of certain hazardous substances in electrical and electronic equipment regulations 2012 (s.i. 2012/3032).” [Online]. Available: https://assets.publishing.service.gov.uk/media/5fcf8c02d3bf7f5d08464d08/ds-0037-21-rohs-notice.pdf

[19] Google, “Plastic free packaging design guide.” [Online]. Available: https://www.gstatic.com/gumdrop/sustainability/plastic-free-packaging-design-guide.pdf

[20] F. S. C. UK, “Recycled materials.” [Online]. Available: https://uk.fsc.org/what-is-fsc/recycled-materials

[21] D. Bourne, “Our next steps on the journey to sustainable hardware.” [Online]. Available: https://blog.google/outreach-initiatives/sustainability/our-next-steps-journey-sustainable-hardware/

[22] Google, “Partnering with suppliers to create better recycled plastic.” [Online]. Available: https://sustainability.google/operating-sustainably/stories/plastics/